

FUNDAMENTALS OF AGRICULTURAL EXTENSION EDUCATION



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BIHAR AGRICULTURAL UNIVERSITY SABOUR BHAGALPUR

Compiled BY

Mr. S. Divakar

Mr. Anil Paswan &

Dr. C.K. Panda

BIHAR AGRICULTURAL COLLEGE,
SABOUR BHAGALPUR-813210 BIHAR
AGRICULTURAL UNIVERSITY, BIHAR

SN	Tittle	Page No.
Chapter 1	Education – meaning, definition and types	4
Chapter 2	Extension education- meaning, definition and process; objectives and principles of extension education	5
Chapter 3	Extension efforts in pre independence era	14
Chapter 4	Extension efforts in post-independence era	17
Chapter 5	Agricultural Development Programe	18
Chapter 6	New Trends In Extension –Privatization	24
Chapter 7	Rural Development	26
Chapter 8	Community Development (CD)	27
Chapter 9	Extension Programme Planning	35
Chapter 10	Communication	41
Chapter 11	Agricultural journalism	60
Chapter 12	Diffusion and adoption of innovation	63

Chapter 1

EDUCATION

Education – It is the process of giving training or instructions to people to develop their knowledge, abilities, skills, characters and mental power.

In other words, it may be defined as the process of production of desirable change in human behavior (knowledge, attitude & skills).

Types of Education

1. Informal Education
2. Formal Education
3. Non-Formal education

Informal Education

It is a lifelong process by which every individual acquires & accumulates knowledge, skills, attitudes and insights from daily experiences & exposure to environment at home, at play etc. **Formal Education**

It is highly institutionalized, uniform, full time, chronologically graded and hierarchically structured 'education system', spanning lower primary school and the upper reaches of the university.

Non – Formal Education

It is an organized, systematic, educational activity carried on outside the framework of the formal system to provide selected types of learning to particular sub groups in population, adults as well as children according to their needs. E.g. Training of farmers.

Difference between Extension Education and Formal Education-

Sl.No	Formal Education	Extension Education
1.	Teaching is largely confined to the premises of the institution	It is largely outside the four walls of the institution.
2.	Learners are homogeneous with common goals	Learners are heterogeneous and have diverse goals.
3.	There is a fixed curriculum, students are examined and degrees are awarded.	No fixed curriculum, it is flexible depending on the need of the learners. No examinations are conducted and no degrees are awarded.
4.	Knowledge flows from teacher to the learners (Vertical)	The extension worker also learns from those whom he teaches (Horizontal). He teaches through local leaders.
5.	Approach is from principles to	Approach is from problem to principles.

	Problem	
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Chapter 2

EXTENSION EDUCATION

The word “*Extension*” was first time used in Britain in 1840 in the form of “*University Extension*” or “*Extension of the University*”. In 1850, William Sewell of Britain used it in his report entitled “*suggestions for the extension of the university*”. But this word was first used in a practical way by James Stuart of Trinity College, Cambridge University in 1867-68, when he was addressing “*Women Association and Working Men’s Clubs*” of North England. Again in 1871, he appealed to the officials of the Cambridge University to establish extension centres under the supervision of university, so that these centres can be used as a platform to deliver lecture to the people. His effort in this field was responsible for him being called “*Father of University Extension*”. In a formal way, Cambridge University in 1873, London University in 1876 and Oxford University in 1878 developed extension system, which worked among people. In 1880, his work came to be known as “*Extension movement*”. Actually the term ‘extension education’ was first introduced in 1873 by Cambridge University. In the eighth decade of the 19th century, the word Extension was used in the form of Extension education in U.S.A. The reason was that in the latter half of the 19th century rural people had started migrating to urban areas. This led to two major problems in U.S.A.:

- (i) due to the migration of rural people to cities, the agriculture suffered a setback, and
- (ii) the increase of urban population created education, employment and housing related problems.

Extension Education is an applied science consisting of content derived from research, accumulated field experiences and relevant principles drawn from the behavioural sciences synthesized with useful technology into a body of philosophy, principles, content and methods focused on the problems of out-of-school education for adults and youth.

J.P. Leagans

,1961. Extension work is an out of school system of education in which adult and young people learn by doing. It is partnership between the Govt. and the people, which provides service and education designed to meet the people. Its fundamental objective is the development of the people.

– Kelsey and Harne

(1963) Extension is a programme and a process of helping village people to help themselves, increase their production and to raise their general standard of living. - D. Ensminger 1967 Extension Education is defined as an educational process to provide knowledge to the rural people about the improved practices in a convincing manner and help them to take decision within their specific local conditions.

– O. P. Dahama

(1973)

Agricultural extension is a professional method of non-formal education aimed at inducing behavioural changes in the farmers for increasing their income through increased production and productivity by establishing firm linkages with research for solving farmer's problems ensuring adequate and timely supply of inputs and using proven methods of communication for speeding of the process of diffusion and adoption of innovations." Y. C. Sanoria (1986)

Extension Education is a science which deals with the creation, transmission and application of knowledge designed to bring about planned changes in the behavior-complex of people; with a view to help them live better by learning the ways of improving their vocations, enterprises and institutions. Reddy A Adivi 1993

The word 'extension' is derived from the Latin roots 'ex' meaning 'out' and 'tensio' meaning 'stretching'. Extension is that type of education which is stretched out to people in the rural areas far and near, beyond the limits of the educational institutions to which the formal type of education is usually confined.

G.L. Ray, 2001

Objectives of Extension

Objectives are the expression of ends towards which our efforts are directed.

The fundamental objective of extension is development of man or

Destination man Specific objectives

- ✓ To assist people to discover and analyze their problems and identify the felt needs.
- ✓ To develop leadership among people and help them in organizing groups to solve their problem.
- ✓ To disseminate research information of economic and practical importance in a way people would be able to understand and use.
- ✓ To assist people in mobilizing and utilizing the resources which they have and which they need from outside.
- ✓ To collect and transmit feedback information for solving management problems.

Philosophy of Extension

Philosophy of Extension Education: The word philosophy is derived from two Greek words 'Philos' and 'Sophia'. 'Philos' means knowledge and 'Sophia' means manner. In this way the correct meaning of philosophy is the manner of achieving the knowledge. Extension education is beneficial aspect of life, which besides being organized, calm and peaceful is welfare oriented. Philosophy of extension education encourages a person to bring about his own development and that of society through his own leadership and motivation by following scientific approach and democratic ways. It further states that the interest of the community should not suffer because of personal interest. In other words philosophy of extension education considers development and progress of individual as a foundation for the development and prosperity of the family, society and the country.

D. Ensminger (1961) has described the philosophy of Extension in the following ways:

1. It is an educational process. Extension wants to change the knowledge, attitudes, understanding and skill of the people.
2. Extension is working with men, women, youths, boys and girls to answer their needs and their wants. Extension is teaching people what to want and ways to satisfy their wants.
3. Extension is “helping people to help themselves.”
4. Extension is working on the basis of learning by doing and seeing.
5. Extension wants development of individuals, their leaders, their society and their world as a whole.
6. Extension is working together to expand the welfare and happiness of the people.
7. Extension is working in harmony with the culture of the people.
8. Extension is a two-way channel process.
9. Extension is a continuous educational process.

Concepts of Extension

- A. Education
- B. Extension Education Process
- C. Salesmanship

Education

It is the production of desirable change in human behavior. Three kinds of behavioral change is produced through extension Education-

- ✓ Knowledge
- ✓ Skill
- ✓ Attitude

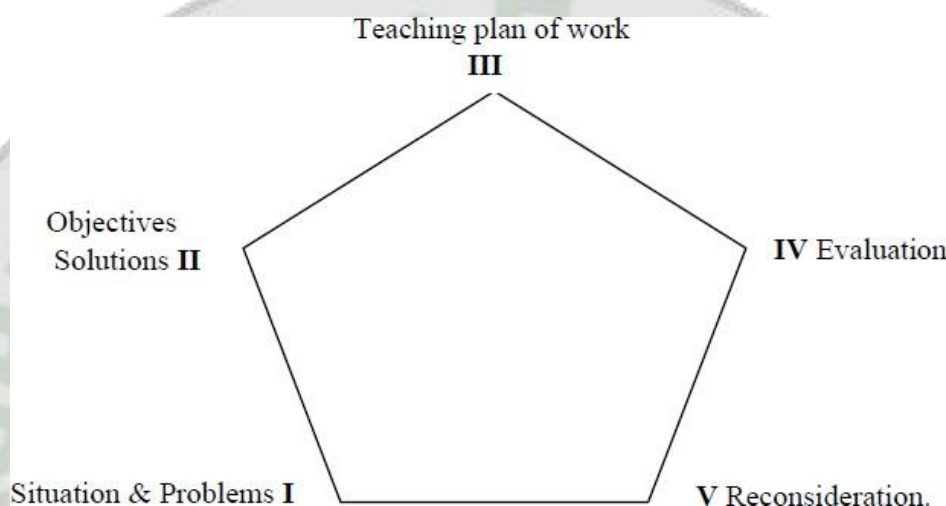
Extension Education Process

The earlier concept of Education aiming at improvement in Knowledge, Attitude, Skills happens step wise in a cyclic manner which is termed as Extension Education Process. The concept of Extension Education Process was developed by Dr. J. Paul Leagans. According to him, Extension Education takes place as a process. The Education starts with study of present situation and identification of problems. Based on the problems identified, solutions are found out and objectives are formulated. In the third phase, plan of work is taught on how to attain the objectives, then the entire process is evaluated and tested whether formulated objectives are attained or not and in the final stage reconsideration about the unattained objectives and unaddressed problems is done. Any Extension work undertaken in villages follows the same path.

I. The first phase is analysis of the situation. This requires a large amount of facts about all aspects of the situation. Facts are needed about the people, their interests (Agriculture, or Poultry or Ag + Horti) education (Eg: literate : 21%, illiterate: 76%, semiliterate; 3%), what they think they need, their social customs, habits and folkways. Facts are needed about the physical situation such as soils Eg : Red soils (29%), black soils(50%), type of farming(Agriculture, or Poultry or Ag + Horti), markets (Mandi or regulated market), size of

farms (small, medium, marginal or big), cropping systems (Paddy fallow pulses or Maize, Maize- Groundnut), housing condition (kuccha or pucca houses), community services and communications.

II. The second phase is deciding upon objectives. It is necessary to distinguish between levels of objectives by separating the general objectives from the specific or working objectives. The planning process must enable the people to select a limited number of problems and to state their objectives clearly. The solutions to be offered must give satisfaction. Objectives should express the behavioural changes in people as well as the economic or social outcomes which are desired.



III. The third phase is teaching. This involves choosing (1) the content or what is to be taught (Pest Management or fertilizer management or organic farming etc) and (2) methods and techniques of communication (Lecture or Method demonstration or Exposure visit). We must use from six to nine different methods of communication to stimulate learning.

IV. The fourth phase in evaluation of the teaching. This should determine to what extent objectives have been reached. This will also be a test of how accurately and clearly the objectives have been stated. The process of evaluation may be simple and informal or it may be formal and very complex.

V. The fifth phase is reconsideration after evaluation has taken place. This step consists of a review of previous efforts and results which reveal a new situation. If this new situation shows the need for further work, then the whole process may begin again, with new or modified objectives. Hence this process is continuous.

Salesmanship

Extension teaching is sometimes compared to commercial salesmanship. It is pointed out that the extension workers are primarily engaged in the sharing of ideas. Normal salesman works for profit but extension personnel work for service.

Functions of Extension:

Change in knowledge - means change in what people know. For example, farmers who did not know of a recent HYV crop came to know of it through participation in extension programmes. The Extension Agents (EAs) who did not know of Information Technology (IT) came to know of them after attending a training course.

Change in skill - is change in the technique of doing things. The farmers learnt the technique of growing the HYV crop which they did not know earlier. The EAs learnt the skill of using IT.

Change in attitude - involves change in the feeling or reaction towards certain things. The farmers developed a favourable attitude towards the HYV crop. The EAs developed a favourable feeling about the use of IT in extension programme.

Change in understanding - means change in comprehension. The farmers realized the importance of the HYV crop in their farming system and the extent to which it was economically profitable and desirable, in comparison to the existing crop variety. The EAs understood the use of IT and the extent to which these would make extension work more effective.

Change in goal - is the distance in any given direction one is expected to go during a given period of time. The extent to which the farmers raised their goal in crop production, say, increasing crop yield in a particular season by five quintals per hectare by cultivating the HYV crop. The EAs set their goal of getting an improved practice adopted by the farmers within a certain period of time by using IT.

Change in action - means change in performance or doing things. The farmers who did not cultivate the HYV crop earlier cultivated it. The EAs who earlier did not use IT in their extension programmes started using them.

Change in confidence - involves change in self-reliance. Farmers felt sure that they have the ability of raising crop yield. The EAs developed faith on their ability to do better extension work. The development of confidence or self-reliance is the solid foundation for making progress.

Principles of extension

Principles are generalized guidelines which form the basis for decision and action in a consistent way.

1. Principle of cultural difference

Culture simply means social heritage. There is a cultural difference between the extension agents and the farmers. Difference exists between groups of farmers also. The difference may

be in their habits,, customs, values, attitudes and way of life. Extension work, to be successful, must be carried out in harmony with the cultural pattern of people.

2. Grass root principle

Extension programmes should start with the local groups, local situation and local problems. It must fit to the local conditions. Extension works should start with where people are and what they have. Change should start from the existing situation.

3. Principle of Indigenous knowledge

People everywhere have indigenous knowledge systems which they have developed through generations of work experienced and problem solving in their own specific situations. The indigenous knowledge systems encompass all aspects of life and people consider it essential for their survival instead of ignoring the indigenous knowledge systems as outdated, the extension agent should try to understand them and their ramifications in the life of the people, before proceeding to recommend something new to them.

4. Principles of interest and needs

Peoples interest and people needs are the starting point of extension work. To identify the real interests and needs of the people are challenging tasks. Extension agents should not pass on their own needs and interests as those of the people. Extension work shall be successful only when it is based on the interests and needs of the people as they see them.

5. Principle of learning by doing

Learning remains far from perfect, unless people get involved in actually doing the work. Learning by doing is the most effective in changing people's behavior. This develops confidence as it involves maximum number of sensory organs. People should learn what to do, why to do, how to do and with what result.

6. Principle of participation

Most people of the village community should willingly cooperate and participate in identifying the problems, planning of projects for solving the problems and implementing the projects in getting the desired results. It has been the experience of many countries that people become dynamic if they take decisions concerning their own affairs, exercise responsibilities for, and are helped to carry out projects in their own areas. The participation of the people is of fundamental importance for the success of an extension programme. People must share in developing and implementing the programme and feel that it is their own programme.

7. Family principle

Family is the primary unit of society. The target for extension work should, therefore, be the family. That is, developing the family as a whole, economically and socially. Not only the farmers, the farm women and farm youth are also to be involved in the extension programmes.

8. Principle of leadership

Identifying different types of leaders and working through them is essential in extension. Local leaders are the custodians of local through and action. The involvement of local leaders legitimization by them are essential for the success of a programme. Leadership traits are to be developed in the people so that they of their own shall seek change from less desirable to more desirable situation. The leaders may be trained and developed to acts as carries of change in the villages.

9. Principle adaptability

Extension work and extension teaching methods must be flexible and adapted to suit the local conditions. It is necessary because the people, their situation, their resources and constraints vary from place to place and time to time.

10. Principle of satisfaction

The end product of extension work should produce satisfying results for the people. Satisfying results reinforce learning and motivate people to seek further improvement.

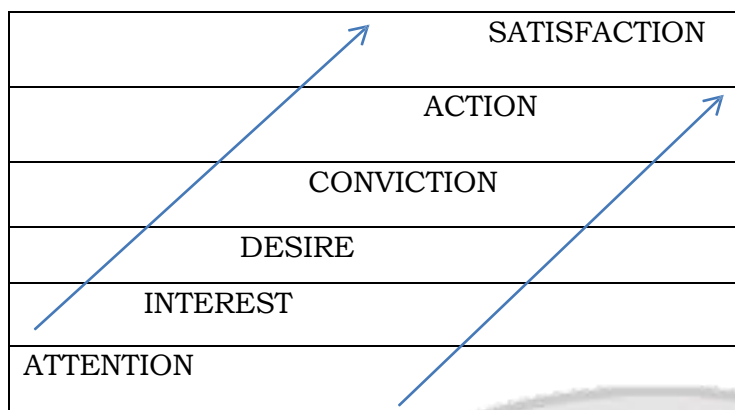
11. Principle of evaluation

Evaluation prevents stagnation. There should be a continuous built-in method of finding out the extent to which the results obtained are in agreement with the objectives fixed earlier. Evaluation should indicate the gaps and steps to be taken for further improvement.

Steps in Extension Teaching

Extension teaching is a planned and deliberate act on the part of the extension worker. The extension worker has to move step by step in a scientific and logical way to impact training to the clients who are farmers, farm women and rural youth. The role of the extension worker is that of a facilitator and motivator. Though details of the procedure may vary from situation to situation.

There are some steps which are basic to extension teaching. These are presented following by Wilson and Gallup (1955).



Steps in Extension Teaching

1. Attention: The first step in extension teaching is to make the people aware of the new ideas and practices. The people must first know that a new idea, practice or object exists. This is the starting point for change. Until the individual's attention have been focused on the change that is considered desired, there is no recognition of a problem to be solved or a want to be satisfied.

Mass methods like Radio, Television, Exhibition etc. and personal contact by the Extension workers, contact through Local Leaders are important at this stage.

2. Interest: Once the people have come to know of the new idea, the new step is to stimulate their interest. This may be done by furnishing them more information about the topic in a way they will be able to understand and use. It is necessary to present one idea at a time relevant to their needs.

Personal contact by the extension worker, contact through local leaders, farm publications, radio, television etc. are important at this stage.

3. Desire: It means un-freezing the existing behaviour and motivating the people for change. At this stage it is necessary to emphasize on the advantages of the new idea or practice.

Visit to demonstrations, farm publications, personal contact by the extension worker, group discussion etc. are important at this stage.

4. Conviction: It is a stage of strong persuasion so as to convince the people about the applicability of the new idea or practice in their own situation and that it would be beneficial for them. The people are furnished with adequate information about the idea and how it works. Field day or Field Visit, Slide Show, personal Contact by the Extension worker and Trainings are important at this stage.

5. Action: This is the stage of the idea or practice into operation. Small scale demonstration with supply of critical inputs may be set up in real life situation

forward. This provides the opportunity of direct experience on the part of the learners. At this stage it is necessary to collect evidence of change such as change in yield, income, employment etc.

Demonstration, personal contact by the extension worker, supply of critical inputs and ensuring essential services are important at this stage.

6. Satisfaction: To produce lasting change, the extension efforts should produce satisfying results. Satisfaction may come from high yield, more income, better health etc. Satisfaction reinforces learning and develops confidence, which generates motivation for further change. To sustain the changed behavior, it is necessary to furnish new and relevant information about the practice on the continued basis till change in the practice itself is felt necessary.

Use of mass media, local leaders and personal contact by the extension workers are important at this stage. Availability of critical inputs and essential services are also to be ensured.

The steps in Extension Teaching are to be synchronized with the innovation-decision process of the learners. The teaching methods and aids mentioned here are by way of example.

Chapter 3

EXTENSION EFFORTS IN PRE-INDEPENDENCE ERA

Characteristics of development programmers' of pre independent era

1. Most of the efforts were based on individual initiatives.
2. Government backing and financing were not forthcoming
3. All attempts were isolated, uneven and discontinuous
4. Staffs were mostly inefficient and untrained
5. No evaluation was carried out
6. No coordination among department
7. Limited involvement of people in programme planning

Sriniketan Project

In 1908, Shri Rabindranath Tagore, under his Scheme of Rural Development work, started Youth Organizations in the villages in the KaligramPargana of his Zamindari(West Bengal). He tried to create the field workers who could identify themselves with the people. In 1921, he established a Rural Reconstruction Institute at Shantiniketan, in collaboration with Mr.

L.K. Elmhirst. A group of eight villages was the center of the programme.

Objectives:

1. To create a real interest in people for rural welfare work.
2. To study rural problems and to translate conclusions into action.
3. To help villagers develop their resources.
4. To improve village sanitation.

Methods to Achieve the Objectives:

1. Creating a spirit of self help.
2. Developing village leadership.
3. Organizing village scouts called "BratiBalika".

Activities:

1. Demonstrations on farmers' fields.
2. Dairy to supply pure milk and animals to farmers or breeding.
3. Poultry farm.
4. Training and organizing the weavers.
5. Training in tanning, pottery, embroidery, tailoring etc.
6. Film shows, meetings, village meals.

Short Comings:

1. Too much emphasis on 'Center'.
2. Confined to limited villages.

GURGAON EXPERIMENT (HARYANA):

Rural uplift movement on a mass scale was first started by Mr. F.I. Brayne, Dy Commissioner in the Gurgaon district of Punjab in 1920. The work gathered momentum after 1933 when Mr. Brayne was appointed as Commissioner of Rural Reconstruction in the Punjab. In 1935- 36, the Government of India granted Rs. One crore for the work which acted as a stimulus. After that the work was transferred to the Cooperative Department and Better Living Societies were organized to take up this work in the villages.

Objectives:

1. To increase agricultural production.
2. To stop wastage of money on social and religious functions.
3. To improve healthy standard of the people.
4. To organize welfare programmes.

Activities:

1. Appointment of village guides.
2. Propaganda through films, folk songs, dramas etc.
3. Rural Economics and domestic Economics Schools.

Short Comings:

1. A one man show.
2. Village guides were un-experienced and untrained.
3. No comprehensive planning.
4. No continuity in the work.
5. Limited to few villages.
6. Force not persuasion.

✓ Started by : F.L. Brayne in 1927 in Gurgaon, Haryana.

ACTIVITIES / OBJECTIVES:

1. Agricultural development and increasing food production.
2. Health improvement.
3. Village sanitation.
4. Social improvement.
5. Reforms in rural institutions.
6. Emphasis on women education.
7. Coordination and publicity.

SEVAGRAM (GUJARAT):

✓ Started by M.K. Gandhi in 1921 in

Sevagram, Gujarat. **PRINCIPLES:**

1. Selp help.
2. Dignity of labour.
3. Self respect.
4. Truth and non

violence. **ACTIVITY:**

1. Use of khadi clothes.
2. Sanitation programme.

3. Women welfare.
4. Upliftment of backward classes.
5. Primary and adult education.
6. Improvement of the condition of poor people.
7. To make the villagers self-sufficient and self-reliant.

MARTHANDAM PROJECT:

- ✓ Started by: Dr. Spencer Hatch in 1921 at Trivandrum.
- ✓ This programme was made with agreement with YMCA (Young Man Christian Association).

PRINCIPLES:

1. Self-help and cooperation.
2. Opening the demonstration center.
3. Helping people to help in their own work.

FIRKA DEVELOPMENT PROGRAMME: (Government sponsored:)

- ✓ It was launched in 1946.
- ✓ Biggest project before Independence.
- ✓ Launched in

Madras state.

OBJECTIVES:

1. Preparation of short term plans for development of rural communication and water supply.
2. Long term plan to make the area self sufficient to agricultural, irrigational and livestock improvement.
3. Formation of panchayat and organisation of cooperatives.
4. Introduction and development of khadi and cottage industries.
5. All round development of rural people.

Chapter 4

EXTENSION EFFORTS IN POST INDEPENDENCE

ERA ETAWAH PILOT PROJECT:

- ✓ Implemented by Albertmayer in 1948.
- ✓ This project is the fore runner of community development

programme. OBJECTIVES:

1. To develop the mental power of people.
2. To awaken the desire of rural people and to make them laborious.
3. To develop agriculture and animal husbandry.
4. Development of panchayat.
5. To develop self-confidence of people.
6. Village sanitation.
7. To measure the extent of agricultural development in terms of social improvement.
8. To build up a spirit of self-help in villagers.

NILOKHERI PROJECT (HARYANA):

- ✓ This project was started in leadership of S.K. Dey in 1948.
- ✓ This project is also called Majdurmanjil.
- ✓ Main aim was to make provision of rehabilitation to migrants who came after partition.

ACTIVITY:

1. Training on agricultural implements preparation.
2. Cottage industry, Carpentry.

Chapter 5

Agricultural Development Programme Intensive Agriculture District Programme - (1960)

The Intensive Agriculture District Programme (IADP) was launched as the first major experiment in intensive agriculture development in the year 1960.

The inadequate rate of growth of agriculture during 1950's was a matter of serious concern to the government. The steep fall in crop productions in the drought year 1957-58 focused attention on the seriousness of the food situation up substantially. In these circumstances, the government of India invited a team of agriculture experts sponsored by the Ford Foundation to make a careful study of Indian agriculture and make recommendations future actions. The team visited India early in 1959 and submitted its report entitled "India's food crisis and steps to meet it in April that year. The team observed that "India is facing a crisis in food production. The crux of the problem is food enough for the rapid increasing population. The team went on to say that "this target (of 110 million tons) can be achieved if an all-out emergency food production programme is undertaken. The best in Indian agriculture is comparable to the best in other countries, but the average level is unduly low. The task before the country is to develop ways of raising the low average to the higher levels that many Indian cultivators have achieved.

The Government of India accepted in general, the recommendations made by this team. The IADP popularly known as package programme was launched in three districts in 1960 - 61 and four districts in 1962 - 63. The seven districts were:

1. Thanjavur (Tamil Nadu)
2. West Godavari (Andhra Pradesh)
3. Shahabad (Bihar)
4. Raipur (Madhya Pradesh)
5. Aligarh (Uttar Pradesh)
6. Ludhiana (Punjab)
7. Pali (Rajasthan)

In the second phase the programme was extended to nine more districts by 1963 - 64. In addition to these, this programme has now been extended to 12 more districts in the country and is in operation in 28 districts in all.

Objective:

The major objectives of the programme were to demonstrate the most effective ways of achieving rapid and significant increase in agriculture production through adoption of a package of improved practices and to suggest successful and adoptable innovations, approaches and procedures which could be adopted in similar areas of the country.

The programme envisaged the selection of favourable areas maximum irrigation facilities and minimum of natural hazards, providing simultaneously all the essential elements of production such as adequate supply of fertilizers, credit etc. It aims at an integrated and intensified approach to the problem of agricultural production in areas which are more responsive to such production efforts. As its immediate goal, the programme sought to achieve rapid increase

in the level of agricultural production through a concentration of financial technical, extension and administrative resources. In the long run it aimed at a self - generating breakthrough in productivity and raising the production potential by stimulating the human and physical process of change.

As this programme demanded more intensive work, the number of village level extension workers was doubled in the community development blocks in the IADP districts. Along with the number of extension officers, the cooperative staff were also increased at the block and district levels.

IADP was a pace - sitting demonstrational programme which showed how to increase production more rapidly on one hand and a path finding experimental programme which facilitated the development of new innovations, new ideas and procedures for wider adoption in agriculture development.

It had covered all the important cash crops grown in the districts although emphasis was to be laid on the major food - grain crops such as paddy, wheat and millets

While the role of the central government was to provide overall direction, guidance and coordination, the programme was of the state government financed by the ford foundation through the central government now the programmed is financed by state government in 28 districts where it is in operation.

High Yielding Varieties Programme(1966):

While both the IADP and IAAP were concerned with the promotion of intensive agriculture, they operated with in the limitations sit by the existing crop varieties which have low fertilizer response. In the year 1963 - 64 a few Mexican dwarf wheat varieties were tried out on selected basis and in 1964 - 65 an exotic variety of paddy, namely, Taichung native - 1 was tried and the results were quite encouraging. Hybrid maize, hybrid bajra and several varieties of improved millets were also in advance stage of development. Though traditional varieties has experienced a moderate increase in yields as a result of the package of practice s emphasized by IADP, the IADP at the same time highlighted the inadequacy of the traditional varieties for the new technology and the need for experimenting with exotic varieties. The droughts of 1965-66 and 1966-67 added greater urgency and importance to this the draft outline of the fourth plan, therefore, inter alia stressed the need for evolving a new approach for boosting up agricultural production over a short span of time, and in 1965 a new programme was instituted cultivation of high yielding varieties which was known as the high

- yielding variety programme. The programme was to be taken up in a phased manner by the state government.

The central government's role was not only limited to evolving and processing the original scheme but also included the coordination at the central level, brining together the state governments, obtaining their consensus and ensuring their participation in the programme operation wise, the state governments were to take the following steps:

1. Selection of areas in each state for the implementation of HYVP.
A phased programme of coverage of HYVP was to be prepared.
2. Development of staff as per with the IADP pattern in the selected districts.
3. Working out of training programme for all levels.
4. Proper arrangements of inputs like fertilizers seeds, pesticides, plant protection equipment and credit on the basis of proper need assessment.

The HYVP was the responsibility of the state government for its implementation from its existing resources under the guidance and supervision of the central government. No separate funds were allocated for this at any level. The area under high yielding varieties increased from six million hectares in 1967 - 68 to 34 million hectares in 1967 - 77. The crop wise coverage in

Crop	Coverage (in million hectares)
Rice	13.5
Wheat	14.7
Maize	1.2
Jowar	2.4
Bajra	2.2

During the year 1977-78 the crop area under this programme was 37.00 million hectares. A target of 40 million hectares has been fixed for 1978-79 further; a target of 52 million hectares has been fixed for 1982 - 83.

The mini - kit programme for rice was started in 1971 and extended for wheat maize and millets in 1974-75, in order to achieve the following objective:

1. Quick popularization of the latest released and pre - released varieties of rice, wheat and millets.
2. To get the farmer's reaction towards such varieties before they taken up for large scale cultivation.
3. Under this scheme, small quantities of seeds of new varieties are made available from research station to farmers, free of cost for trial purposes.

Institution Village Linkage Programme (IVLP)(1995)

Technology Assessment and refinement through Institution Village Linkage Programme (IVLP) is one of the recent programmes of the ICAR emphasizing on technology transfer, refers to the process or a set of activities that must happen it new scientific agricultural information is put to use in any farm production system. In view of three districts types of agriculture, viz. commercial green revolution and complex, diverse and technology assessment and refinement with the emphasis on stability and profitability with environment concerns for well-defined farm production system and appropriate for value addition in case of commercial for production system. Thus, IVLP is a programme for generating appropriate technologies by refining and assessing innovation generated by scientists knowledgeable about their respective disciplines however, various factors operating in the farm production system of small farmers hitherto not considered by scientists, affect adoption process. IVIP, therefore, has following specific objectives:

1. To introduce technological intervention with emphasis on stability and sustainability along with productivity of small production system.
2. To introduce and integrate the appropriate technologies to sustain technological interventions and their integration to maintain productivity and profitability taking environmental issues into considerations in comparatively well-defined production systems.
3. To introduce and integrate the appropriate technologies to increase the agriculture productivity with marketable surplus in commercial farm production systems.
4. To facilitate adoption of appropriate post - harvest technologies for conservation and on - farm value addition of agriculture products, by products and wastes for greater economic dividend and national priorities.
5. To facilitate adoption of appropriate technologies for removal of drudgery increased efficiency and higher income of farm woman.
6. To monitor socio-economic impact of the technologies intervention for different production systems.
7. To identity extrapolation domains for new technology modules based on environmental characterization at meso and mega levels.

The IVLP is being implemented through ICAR institute headquarters, SAUs headquarters, and Regional Research Stations, Zonal Research Stations of the ICAR institute, SAVs and KVKs. The centers selected for the purpose have required infrastructural facilities, viz. in terms of team of scientists, laboratory, transport; each centre has to adopt a village or cluster of villages representing of the district.

National Agriculture Technology Project (NATP) (1999)

After the WTO, there was rapid global change in agricultural trade. There are emerging needs to radically reform the extension system to meet the challenges of rural development. Therefore, National Agriculture Technology Project was conceived as a pilot project. It was launched in November, 1998. It was an initiation of government of India with World Bank Assistance. Its purpose was to make the extension serious broad based, holistic beyond agriculture technology transfer besides to consolidate earlier, weakness and gaps.

Objectives -

The overall objective of NATP was to revitalize the agriculture technology generation assessment, refinement and dissemination systems. It had three components, there were: -

- ICAR organization and management reforms
- Expansion of location specific research on production system and.
- Testing of innovations in technology dissemination.

The innovation in technology dissemination (ITD) component of the project aimed at addressing key constraints in technology generation, validation and dissemination by introducing new institutional models and operational reform process in selected states. The states were namely Bihar, Andhra Pradesh, Himachal Pradesh, Jharkhand, Maharashtra, Orissa and Punjab. The project was pilot tested in four districts of above mentioned each state.

Under NATP, some institutional innovations were made in each project district.

These were as follows:

- Establishment of decentralized district level autonomous institution named ATMA (Agricultural technology management agency)
- Establishing FAIC (Farm Information and Advisory Centre at block level).
- Establishment of state level agricultural extension management training institutions named SAMETI (State Agricultural Extension management and training institutes).
- Evolutions of state levels coordination and policy making body named IDWG (Inter Department Working Group)

With an objective to address the expanding role of extension certain operational reforms processes were initiated in the project districts like strategic planning, bottom-up planning, promotion of farmers' interest group and farmers. Organization, farming system approach decentralized decision making farmer participation research market led production sequential extension interventions, cost and resource sharing IT and communication support, public private partnership, techno managerial focus team approach devitalizing ongoing schemes and farmers focused and farmers accountable extension based on the performance of the pilot study ATMA model is now introduced in various states of the country with the help of government of India.

Agricultural Technology Management Agency (ATMA)(1999)

ATMA is a registered society responsible for technology dissemination at district level. It is a focal point for integrating research extension and

expanding project funds entering into contracts and agreements and maintaining revolving accounts. It has also linkage with all the line department research organizations nongovernmental organizations and organizations associated with agricultural development in the districts.

Each ATMA has a governing board. The district collector is the chairman of the board of the ATMA with other district level officials such as agricultural veterinary fishery etc as member the farmers woman representative are also taken ability to take all discussion concerning programme planning resource allocation.

The key functions of ATMA governing board are as follow (MANAGE. 2004)

- Review and approve strategic research and extension plan and annual action plan for the district.
- Receive review annual reports presented by the participating unit and to provide feedback and direction to them.
- Receive and allocate project funds to carry out projects research and extension activities.
- Foster the organization and development of farmers' organization and farmer interest group.
- Facilitate the greater involvement of private sector.
- Encourage agriculture credit institutions to increase the availability to capital to resource poor farmers.
- Arrange for the periodic audit of ATMA financial accounts.
- Adopt and assess the rules and by laws for the ATMA.

Besides 'governing board' ATMA has a management committee. It is the executive arm of ATMA. It implements the project through this block level outfit. The district level official of department of agricultural, horticultural animal husbandry etc. and heads of Zonal Research Station and KVKs of the district are the official members of the committee. There are also NGO and farmers representatives in the committee. The key farmers representatives in the committee. The key functions of the committee are to carryout periodic PRA (Participatory Rural Appraisal), to identify the problems and constraints faced by different socio - economic groups and farmers, to prepare SREP and annual action plan for the district, to coordinate execution of annual action plan, to establish coordinating mechanisms at the block levels etc. **Agricultural Technology Information Centre (ATIC)**

The Agricultural Technology Information Centre (ATIC) is a "single window" support system linking the various units of a research institution with intermediary users and end users (farmers) in decision making and problem-solving exercise.

The cornerstone of India's agricultural revolution has been the availability of improved varieties of cereals, oilseeds, pulses, etc. breed of livestock including poultry and fisheries; horticultural plant materials, and improved management practice for increase productivity, sustainability and stability of various crops and livestock enterprises. This has raised the search by farmers for future availability of seed, planting materials and other materials, easy accessibility to diagnostic services for soil fertility and plant protection, availability of appropriate information through leaflets and pamphlets and increased scope in sale of consultancy services. Often the farmers are not aware as to whom and where to approach for field problems. It is felt that the facility of a '**single window**' approach at the entrance of the ICAR Institute/ State Agricultural Universities will enable the farmers to have the required information for the solution to their problems related to the areas in which the concerned institute is involved. Because of the dominance of small and resource poor farmers and concentration of poor people in several sectors, public institutions like ICAR institutes and SAU's will continue to play a vital role in supply of information for increasing the overall productivity in agriculture.

The rationale for establishment of ATIC are:

- i. to provide diagnostic services for soil and water testing, plant and livestock health;

- ii. to supply research products such as seeds and other planting materials, poultry strains, livestock breeds, fish seed, processed products, etc, emerging from the institution for testing and adaptation by various clientele;
- iii. Providing information through published literature and communication materials as well as audio-visual aids; and
- iv. Providing an opportunity to the institutes/ SAU's to generate some resource through the sale of their technologies.

Objectives:

The objectives for establishment of such centers as a single window system are as follows:

- To provide a 'single window' delivery system for the products and species available from an institution to the farmers and other interested groups as process of innovativeness in Technology Dissemination at the institute level.
- To facilitate direct the farmers access to the institutional resources available in terms of technology, advice, technology products, etc. for reducing technology dissemination losses; and
- To provide mechanism for feedback from the users to the institute.

All the information on technologies, techniques, knowledge and materials, including planting materials, seeds, advisory services, diagnostic services, etc are proposed to be supplied through a single window system. This requires a small building - to be provided/constructed/renovated by the host institute- at the entrance of the main campus so that the ATIC window is easily accessible to the farmers, state development officials and other users. This structure should have a sale counter, a hall for screening the technology on TV /VCR, technical books/ an information room, provision of tea/coffee, attached toilets and open space for keeping planting materials/varieties. In order to keep track of the progress in the implementation of the programme approved for funding the reporting schedule will be in the form of half-yearly, annual and completion reports, prescribed proformas for which will be provided. The schedule of the reporting and submission of audited accounts are to be strictly followed.

Obligation of the participatory institution.

The participating institutions/ SAUs are expected to provide support in terms of technological as well as methodological backstopping for implementation of the Project. The participating institution/SAU shall undertake:

- To provide the basic infrastructure facilities for the ATIC;
- To provide the salaries and other facilities and services necessary for implementation of the project.
- To assist in the implementation and the monitoring of the project. The full cost of this element shall be borne by the participating institution.
- To ensure timely preparation and accuracy of accounts and their audit.
- To follow up the prescribed procedures for procurement of stores and services etc. as guided by NATP or World Bank.

Certificate of undertaking to this effect is to be given by the participating institutions/SAUs.

Chapter 6

New Trends in Extension –Privatization

Extension has been, and still is, under attack from a wide spectrum of politicians and economists over its cost and financing. As a result, Extension Systems have had to make changes, by restating the system's mission, developing a new vision for the future, and formulating plans for the necessary transition to achieve the desired change.

At least three scenarios have been suggested by government and farm organizations with regard to privatization of extension:

1. Public financing by the taxpayer only for the kinds of services of direct concern to the general public.
2. Direct charging for some individual services that produce direct return in the form of improved income, with the possibility of differential rates for specific situations or target groups.
3. Mixed funding shared between public and private professional association contributions for services, with delayed return or collective services, such as applied research, training of farmers and agents, and improvement in Extension methods and tools.

Extension service has been traditionally organized and delivered by the public sector all over the world, which led to a situation wherein, whenever one refers to extension, it denoted public extension service. Similarly, whenever private sector is referred to, there is a tendency to consider only the corporate sector in the category. However, private extension has a broader canvas including all relevant private groups than the narrow canvas of corporate sector.

Privatization of extension services does not aim at substituting private sector for public extension service. In fact, privatization has adopted a variety of forms involving different stakeholders. The paper portrays the major stakeholders, viz, private corporate firms, credit institutions, farmer's associations non-governmental organizations and media organizations and Government organizations and analyses their participant configurations.

The success of an extension service depends on the effectiveness of planning at four levels policy, programmes, projects and strategy. Policy and programmes must be decided by the public extension system, while projects and strategy can be formulated by the private extension organizations. When the private extension organizations get involved in providing extension support to farmers, it is likely there will be competition among the various extension providers, which will result in more efficient and demand-driven service. Both technical and allocative efficiency which are basically economic in nature are well take care of by the private extension agencies, resulting in cost minimization, profit maximisation and optimal use of resources, which are warranted in a competitive environment.

Public extension service often views sustainability of programmes only in terms of continuity. Sustainability is different form continuity, which has both ecological and equity dimensions. The private extension agencies, especially NGOs and media organizations provide valuable service in ensuring sustainability of programmes in terms of the above two dimensions.

The private extension system in India offers the following services for farmers – terms of sharing, augmenting and supplementing the public extension efforts besides offering unique and innovative initiatives, which the public extension service can also emulate. Some of the Services are:

1. Cost sharing by farmers' groups
2. Cost recovery on selected services offered to farmers
3. Contracting services to small groups
4. Paid extension services for affordable farmers
5. Value addition by agro-processing firms
6. Consultancy services (both technical and managerial)
7. Privatised service centres for farmers
8. Assisting Self Help Groups of farmers
9. Information support through media organizations

Private extension system can offer a variety of services for farmers in a competitive environment which the public sector may not be able to. Hence, it is suggested that public sector extension may limit its activities only to regulatory and enabling functions, and should mainly focus on educational activities, which are unattractive to private sector delivery.

Chapter 7

Rural Development

Rural - Is an area, where the people are engaged in primary industry in the sense that they produce things directly for the first time in cooperation with nature as stated by Srivastava (1961).

Society -A society or community can be classified as rural based on the criteria of lower population density, less social differentiation, less social and spatial mobility, slow rate of social change, etc. Agriculture would be the major occupation of rural area.

Rural Development – It is a process which aims at improving the well being and selfrealisation of people living outside the urbanized areas through collective process. **According to Agarwal (1989)** rural development is a strategy designed to improve the economic and social life of rural poor. The rural development in general terms is a strategy designed to improve economic and social life of people in a rural settlement.

Objectives: The major objectives of rural development are:

- 1) To achieve enhanced production and productivity in rural areas,
- 2) To bring about a greater socio-economic equity,
- 3) To bring about a spatial balance in social and economic development,
- 4) To bring about improvement in the ecological environment so that it may be conducive to growth and happiness, and
- 5) To develop broad based community participation in the process of development.

Scope and Importance of Rural Development

Rural development is a dynamic process which is mainly concerned with the rural areas. These include agricultural growth, putting up of economic and social infrastructure, fair wages as also housing and house sites for the landless, village planning, public health, education and functional literacy, communication etc.

Rural development is a national necessity and has considerable importance in India because of the following reasons.

1. about three-fourth of India's population live in rural areas,
2. Agriculture contributes nearly 14% of GDP.
3. around seventy per cent of Indian population get employment through agriculture,
4. bulk of raw materials for industries come from agriculture and rural sector,
5. increase in industrial population can be justified only in rural populations' motivation and increasing the purchasing power to buy

Chapter 8

Community

Development (CD) Subtopic I:

Genesis of CD

Subtopic II: Meaning and definition of CD

Subtopic III: Assumptions underlying the Philosophy of
Community Development

Subtopic IV: Basic Philosophy

Subtopic V: Objectives of CD

Subtopic VI: Organizational Set-up for CD Extension Service

Subtopic I: Genesis of CD

During the pre-Independence era, various efforts were made towards rural development, in various parts of the country. The Government efforts to bring about improvement in rural areas were directed through fairly well organized development departments, such as agriculture, irrigation, animal husbandry, cooperation, health, education

etc. These however, worked independently of one another without common objective and they used to approach the last official in the villages who is in touch with the villager. The area covered by him was so vast that the villagers hardly felt his presence. The activities were not linked up, with revenue officers of the area, or local bodies etc.

Soon after Independence (1947) the Central Government has initiated Grow More Food Campaign and Etawah project with the objectives of attainment of self-sufficiency in food grains and over all development of the rural areas.

As a result of the lessons and experiences of the Royal Commission on Agriculture, the Grow-More-Food Enquiry Committee and the successful experience of Etawah Project, 15 Pilot Projects were started in 1952 in selected states, with financial assistance received from the Ford Foundation. The Planning Commission which drafted First Five Year Plan gave the highest priority to the development of agriculture and irrigation. The Commission prescribed Community Development as the method for initiating the process of transformation of social and economic life of villages and "Rural Extension" as its agency.

For undertaking this new programme, the Government of India entered into an operational agreement with the Government of U. S.A. under T.C.M. (Technical Cooperation Mission). Fifty five Community Development Projects were started on 2nd Oct, 1952 the birthday of Mahatma Gandhi.

Each project of Community Development Programme consisted of about, 300 villages covering over 400-500 square miles and having a population of around 2 lakhs. The project area was divided into these development blocks, each comprising 100 villages and population around 60 to 70 thousands. These blocks were further divided into groups of 5 to 10 villages, each group in the charge of multi-purpose village level worker.

Subtopic II: Meaning and definition of CD

Community: A community consists of persons in social interaction within a geographical area and having common centers of interest & activities and functioning together in the chief concern of life.

Development: The term development connotes growth or maturation. It implies gradual and sequential phase of changes.

Programme: A programme is the total educational job being done in a particular setting. It is a prospectus or a statement issued to promote understanding and interest in an enterprise.

Development may be described as a

Process of change from the traditional way of living of rural communities to Progressive ways of living;

Method by which people can be assisted to develop themselves on their own Capacity and resources, Programme for accomplishing certain activities in fields concerning the welfare of the rural people and

Movement for progress with a certain emotional and ideological content The two essential elements in community development are

1. Participation by the people themselves in efforts to improve their level of living with as much reliance as possible on their own initiative and
2. The provision of technical and other services in ways which encourage initiative, self-help and mutual help and make these more effective.

According to Taylor(1956) CD is a method by which people in villages are involved in helping to improve their own economic and social condition and thereby they become effective working groups in the programme of their national development. This carries the assumption that people become dynamic if they are permitted to make their own decisions and are helped to carry on the programme.

The Cambridge Conference defining CD – “It is a movement designed to promote better living for the whole community and if possible on the initiative of the community.”

Subtopic III: Assumptions underlying the Philosophy of Community Development

1. Communities of people can develop capacity to deal with their own problems
2. People want change and can change
3. People should participate in making, adjusting or controlling the major changes taking place in their communities
4. Changes in community living that are self-imposed or self-developed have a meaning and permanence that imposed changes do not have
5. A "wholistic approach" can deal successfully with problems, with which a "fragmented approach" cannot cope
6. Democracy requires the people's cooperative participation and action in the affairs of the community
7. Frequently communities of people need help in organizing to deal with their needs

Subtopic IV: Basic Philosophy

1. **Work based on “felt need”-** The programme should help the community to solve some of the problems which it feels are existent.
2. Work based on assumption that people want to be free from poverty and pain.
3. It is assumed that people wish to have freedom in controlling their own lives and deciding the forms of economic, religions, educational and political institutions, under which they will live.
4. **People’s values given due consideration:** It is presumed that cooperation, group decision making, self-initiative, social responsibility, leadership, trustworthiness and ability to work are included in the programme.
5. **Self Help:** The People actually plan and work on the solution of their problems themselves. Amelioration from external forces/agencies always to be need based of the local community concern. **People are the greatest resource:** Through the participation and empowerment of local people they will developed.
6. **Empowering rural women:** A s women contribute 50% of rural community, their empowerment ensure smooth function of CD programme.
7. The CD programme involves changes in knowledge, skill and attitude of rural people.

Subtopic V: Objectives of CD

The fundamental or basic objective of Community Development in India was the development of people. Its broad objectives were: (i) economic development (ii) social justice and (iii) democratic growth.

Basic objectives:

- i. The all-round development of the rural community.
- ii. To develop the feeling of communitarian life style among the rural people
- iii. To develop the feeling of responsibility, to create confidence, to create inspiration for working by self-decision among the rural people and establishing local leadership and institutions this can tackle the problems of that area.

Objectives:

1. To increase the agricultural production
2. Community and integrated development
3. The extension of the new scientific knowledge
4. Development of small and medium irrigation projects
5. Development of co-operative organizations
6. Construction of roads.
7. To increase the adult education and primary education
8. Facility for entertainment.
9. Development and construction of primary health care centre and the public health service. To inspire the youth for the development programme.

Subtopic VI: Organizational Set-up for CD Extension Service

The organizational set-up for Community development Programme runs form the national level through state, district and block levels to the village level and there are three main constituents of this new set-up.

1. The direct-line staff such as State Development Commissioner, B.D.O and

Village Level Worker.

2. The auxiliary or specialist staff, such as different heads of technical departments at the state and district levels and extension officers at the block level.

3. Panchayati Raj System- The Zila Parishads, Block Samitis and Village Panchayats.

A) National Level:

At the National level programme, the policies are formulated by the National Development Council presided over by the Prime Minister of India. Membership of the Council consists of the Central ministers of the concerned ministers, chief ministers of all states, and members of the Planning Commission. The Planning Commission provides guidance for Plan formulation and gives it approval to annual and Five-year Plans of the states as well as of the Centre. The Ministry of Agriculture and Irrigation is responsible for giving national guidance, policy formulation and technical assistance in regard to Agriculture Extension and Community Development (now Rural Development programmes). In the Agriculture department, the Agricultural Commissioner, Government of India, assisted by a number of assistant commissioners and directors, with the supporting staff, is in charge of all agricultural development programmes at the national level. Within this Department, special mention may be made of the Directorate of Extension Training responsible for the training of Extension officers, VLWs, instructors of Village-Level Workers Training Centers and others and the Directorate of Farm Information which is concerned with the dissemination of new agricultural technology and innovations through various media.

B) State Level:

At state level also, there is usually a State Development Committee presided over by the Chief Minister of the state with the other concerned ministers as its members. This Committee is responsible for the state's plan and programmes and for fixing the targets for regions and districts. Besides this committee, there are usually a number of other advisory or technical committees.

As regards the actual administrative functioning the State Development Commissioner is the top-level executive responsible for directing, coordinating and providing overall guidance for development programmes and maintaining a two-way channel of communication between the state governments and the Central government. He co-ordinates the activities of different development departments, such as agriculture, animal husbandry, co-operation, panchayati raj, health, education, irrigation, power and electricity. The heads of these technical departments are responsible for planning and implementing the technical programmes and for providing the necessary technical guidance, manpower and support.

C) District Level:

At the district level also, there is usually a District Development or District Planning Committee presided over by the District Collector or Deputy Commissioner. The other members of this committee are the heads of the departments in the district, chairman and vice-chairman of the district boards, representatives of voluntary organizations, local bodies and members of parliament and state legislatures. In the states, where the Panchayati Raj is operating, the Zila Parishads are responsible for

planning, co-ordination and consolidating the development programme in the district. The district collector is the key official who co-ordinates the activities of all development departments at the districts level. The district-level technical heads of agriculture, animal husbandry, co-operation, panchayats, public health, irrigation, education and rural industries are responsible for planning and implementing the development programmes relating to their departments. Administratively, they are responsible to the district collector on one hand and to their state heads of development departments on the other.

D)BlockLevel:

A district is subdivided into a number of community development programmes. The block development officer is the head of the block team, and co-ordinates all the activities of the development departments at the block level. He is assisted by eight extension officer's from different fields, namely agriculture, animal husbandry, health, co-operation, panchayats, engineering, social education and rural industry. At the non-official level in the states, where the Panchayati Raj has been implemented, the Panchayati samiti (also called the Block), this Samiti) has the statutory powers for formulating and executing development programmes. The Samiti is assisted by the B.D.O. and the extension officers. Wherever the Panchayati Raj is not working, there are block development advisory committees.

E) Village Level:

At the village level, the multi-purpose village-level worker is the main extension staff. He is the last extension functionary in the administrative hierarchy and is the main contact person. He is responsible for all developmental work at the village level, and forms a connecting link between the various technical departments and the rural people. Usually, in a normal community development block, there are 10 village-level workers. Their number has been doubling in the intensive Agricultural Development Programme (IADP) blocks. On the non-official side, usually there is Panchayat in every village or for a cluster of villages, and is responsible for planning and implementing the community development programmes' and ensuring people's participation in them.

Guiding Principles of Community Development

The United Nations Economic and Social Council have been trying to define such principles and concepts of Community Development as well acceptable to all its-member-States and which they should agree to implement as far as possible.

1. Activities undertaken must correspond to the basic needs of the community; projects should be initiated in response to the expressed needs of the people.
2. Local improvements may be achieved through unrelated efforts in each substantive field; however, full and balanced community development requires concerted action and the establishment of multipurpose programmes.
3. Changed attitudes in people are as important as the material achievements of community projects during the initial stages of development.
4. Community development aims at increased and better participation of the people in community affairs, revitalization of existing forms of local government and transition towards effective local administration where it is not yet

functioning.

5. The identification, encouragement and training of local leadership should be a basic objective in any programme.
6. Greater reliance on the participation of women and youth in community projects.
7. To be fully effective, self-help projects for communities require both intensive and extensive assistance by the Government.
8. Implementation of community development programme on a national scale requires adoption of consistent policies, specific administrative arrangements, recruitment and training of personnel, mobilization of local and national resources and organization of research, experimentation and evaluation.
9. The resources of Non-Governmental Organizations should be fully utilized in Community development Programmes at the local, national and international level.
10. Economic and social progress at the local level necessitates parallel development on a wider national scale.



Subtopic VI: Relationship between Extension education**and CD SIMILARITIES BETWEEN EXTENSION EDUCATION****AND CD**

1. Both the processes are essentially educative
2. For both central objectives is the “Growth of people”
3. Both processes are inter disciplinary in chapter
4. Both aim at bringing about change
5. Both are involvement processes
6. Both are relatively slow processes
7. Both are government sponsored and supported organizations.
8. Both emphasize on cooperation
9. Both are concerned with social and economic development

EXTENSION EDUCATION	COMMUNITY DEVELOPMENT
Place emphasis on individual development	Emphasizes cooperative or group action for the benefit of communities
More indirect approach to help people to help themselves	Direct government approach to help people.
Aims to bring about changes by leaving decisions for change of individuals	Decision are taken by group or representatives
Extension education is highly specialized and concentrates on agricultural changes	Concerned with various elements of human concern including health, sanitation etc
Mono purpose approach	Multipurpose approach
Extension education is concerned with improving social organizations but not attack problems at this level	C.D. is directly concerned with developing social organizations
It is the Means	It is the End

Short Note
National Extension
Service

Rural development activities under Government sponsorship was started with the introduction of community development programme on 2nd October 1952. Within a few months of the launching of these pilot projects, it was prominently experience that the people were ready even keen, for the programme. The people in all the project areas responded enthusiastically and indeed much beyond the expectations of the Government and the sponsors of the programme. This fact emphasized the need for a rapid extension of the programme to other parts of the country. But the country's resources were not sufficient to sustain a comprehensive plan of the same magnitude as contemplated in the first 355 projects. The Government therefore decided to launch alongside the community development programme another programme which was somewhat less intensive in character, called the National Extension Service programme. The National Extension Service programme was formulated in April 1953 and it was inaugurated one year after the 55 community projects that is, on October 2, 1953. It was a major development in the sphere of rural reconstruction in India. Since the basic idea underlying both the community development and National Extension Service programme was the same, the two were integrated under one agency at the Centre as well as in the states. Bothe the programmes were complementary and interwoven and ran concurrently. The idea behind the National Extension Service Programme was to cover the entire country within a period of about 10 years, that is to say, by 1960-61.

The inter-relation between the community development programme and Nation Extension Service described as follows: It is necessary to explain the inter-relation between the community development programme and the National Extension service. The movements have identical aims. The National Extension Service is a permanent organization and will cover the whole country. It provides the basic organization, official, non-official and a minimum financial provision for development. Further funds will be found from the central Government and the State's own allotments under different heads. National Extension Service blocks in which successful results have been achieved with the maximum popular co- operation are selected for intensive development for a period of three years. This intensive development will depend on the available financial resources and local support and local support and enthusiasm. The National Extension Service and the community development programmes have uniform unit of operation which is called a development block. It represents on an average 100 villages, with a population of 60,000 to 70,000 persons spread over an area of 150 to 170 square miles. But the N.E.S. blocks are not developed with the same intensity as areas under the community development blocks. Out of the areas developed as National Extension Service Blocks, selection is made periodically for intensive development work under the community development programme and the block which are selected are C.D. blocks. Only those blocks are selected which in their working showed good results and where people's participation had been in abundance.

Chapter 9

EXTENSION PROGRAMME PLANNING

Meaning & definition

The word 'Programme' has several distinct meanings in the dictionary. It means a proclamation, a prospectus, a list of events, a plan of procedures, a course of action prepared or announced beforehand, a logical sequence of operations to be performed in solving a problem.

Extension Programme

According to Kelsey & Hearne (1949), "an extension programme is a statement of situation, objectives, problems and solutions".

Planning: According to Myerson and Banfield (1935) 'Planning is designing a course of action to achieve ends'.

Extension Programme Planning: According to Musgrow (1962) Extension Programme Planning is the process of determining, developing and executing programmes. It is a continuous process, whereby farm people, with the guidance and leadership of the extension personnel, attempt to determine, analyze and solve local problems.

In this there are three characteristics-

- what needs to be done;
- when it should be done; and
- how it should be done

Some related important terminology:

Situation is a brief statement of the more general factual information together with the needs and Desires of the people.

Plan of work is an outline of activities so arranged as to enable efficient execution of the entire programme. It answers the questions how, when, where and by whom the work is to be done.

Project is an outline of procedure and pertains only to some phase of extension work.

Calendar of work is a plan of work arranged chronologically.

Aims are generalized and broad statements of directions with respect to given activities.

Eg: Extension Worker's aim is to improve farmers' economic

condition. Objectives are expressions of ends towards which

our efforts are directed **Eg: To increase paddy yield by 40%**

Goal is the distance in any given direction one expects to go during a given period of time.

Eg: To increase Paddy yield by 20 Q/ha. in the

current year. Plan is a predetermined course of action.

Problem- It is a condition that the people after study, with or without help have decided needs changing

Solution- It is a course of proposed action to change an unsatisfactory condition to one that is more satisfying

Programme development process is defined as a continuous and cooperative activity involving

lay people and the extension staff in which problems are identified, objectives are set forth and action is taken to reach the objectives – **Sanders**

Importance of Programme Planning

1. Helps in careful consideration of what is to be done and why.
2. Helps to furnish a guide or straight edge against which to judge all new proposals can be compared. A good programme guide tends to keep everyone concerned. It keeps them to avoid less important issues.
3. Programme helps to establish objectives towards which progress can be measured and evaluated.
4. Formulation of a programme gives continuity particularly during changes in personnel.
5. Programme helps to aid in the development of leadership.
6. It also serves as a means of choosing: a) the important from the incidental Problems b) the permanent from the temporary changes.
7. It helps to prevent mistaking the means for the end, and to develop both felt and unfelt needs.
8. Programme avoids wastage of time and money and promote general efficiency.
9. Programme helps to co-ordinate the efforts of the different people working for rural development.
10. Programme gives the extension workers more support by local people, thus ensuring cooperation and financial support.
11. Finally it helps to produce a written form of statement for general public use.

Topic- Evaluation and monitoring of Extension Programme

MONITORING

Monitoring is the process of assessing / analyzing progress on going of a project based on predetermined indicators and objectives of programme.

Monitoring simply means to keep a watch on what is happening. Monitoring is generally done at the operational level and involves collection and analysis of information. To be effective, monitoring should be a continuous process.

The monitoring of extension activities consists of timely gathering and analysis of data on the organization of the extension service and its performance, as well as on farmers' acceptance of extension advice.

Monitoring reveals whether the components of the extension service are operating as intended. It is done through survey, baseline, meetings, discussions, reporting, structured interviews, questionnaire surveys etc.

For monitoring indicators are of 2 types – (1) Process indicators, (2) Impact indicators There are qualitative / quantitative indicators. Qualitative indicators use proxy indicators e.g. for measuring leadership ask 2-3 questions indirectly to assess it like how many women can take decision, how many can lead a group on her own, how many women can handle crisis situation.

Quantitative indicators deal with numerical values. They are easy to measure. Qualitative indicators assess changes in knowledge, attitude, behaviour and practices at individual, family, community levels.

EVALUATION IN EXTENSION

Evaluation is a very necessary step for extension activities. About those extension programmes which have been drawn up and introduced, it is necessary to know how much effective they have been or how many new things of agriculture the farmers have learnt from them. If they did not get any advantage, then, what were the causes? What are the reasons for farmers not taking interest? The study of these points is the evaluation in extension.

Definitions of evaluation:

1. Programme evaluation is the determination of extent to which the desired objectives have been attained or the amount of movement that has been made in the desired direction (Boyle and Johns, 1970)
2. Evaluation is the comparison of the situation before and after a development programme has operated with an area for a predetermined period (Metthew, 1956)
3. The process of assessing impact of project / programme *vis-à-vis* set goals / objectives
4. Evaluation in its broadest sense means judging the value of something.
5. Evaluation may be defined as a process of systematic appraisal by which we determine the worth, value or meaning of something. This something in extension may be a programme or part of a programme, a method used in carrying extension work, or a situation such as a community a block or even a larger area.
6. Extension evaluation is the process of determining how well the desired behavioral changes have taken place or taking place because of extension education effort.

Objectives of Evaluation:

1. To know the causes for success or failure of the programme, along with identifying the obstacles for success in the programme
2. To inspire the workers for the evaluation of their objectives
3. To know the merits and demerits of the programme
4. To increase the self-confidence in both the rural people and extension worker
5. To unearth the expenses and achievements of a programme
6. To find out the usefulness of new experimental teaching methods

Types of evaluation:

- I. According to purpose for which we conduct evaluation- it is of two types
 - **Formative evaluation:** gathers data for development of an effective extension programme.
 - **Summative evaluation:** tries to measure end results of a programme in order to decide whether or not it should be continued or discontinued.
- II. According to the stage of the programme- when the evaluation is conducted. It is of two types
 - **Ongoing or process evaluation:** Ongoing or process evaluation means evaluation when the work is in progress, it enables the evaluator and the stakeholders to develop a better understanding of the functioning of the programmes.

- **Outcome evaluation:** Outcome evaluation is done after completion of the work. It enables the participants to assess which of their goals are being achieved and how well this is being done.

III. According to the nature of evaluation- it is divided into 5 degrees as suggested by **Frutchey (1967)** Fig: Evaluation continuum of **degrees of evaluation**

Casual every day	Self- checking	Do-it- yourself	Extension studies	Scientific
evaluations	evaluations	evaluations		research
1	2	3	4	5

The first three degrees of evaluation are informal in nature and the next two degrees of evaluation are formal in nature.

1. Causal every-day evaluation: This is the initial part of evaluation we do every day. Eg- good dress, best actor, worst speech etc. These simple observations are important but have their own limitations. We must be careful in analyzing what is the truth and what is seen. It is most crude and subjective method of evaluation.

The following are some of the limitations of this type of evaluations:

- Personal ideas used instead of standard measurements.
- Intuition (guess) and personal bias cannot be eliminated
- No systematic plan for arriving at conclusion
- May have only part of the information.

2. Self-checking evaluation: This is the next higher degree of evaluation. It makes conscious attempt to apply principles of evaluation. Eg- checking on ordinary observations, talking with others, getting other people's judgments etc
Eg: if person X comes to an extension order and says that in Ramapur village, farmer Jitan Kumar got 100 bags / acre of yield with Y variety, Extension worker instead of directly accepting this information, (s) he personally visits the field of the farmer Jitan Kumar and talks to the villagers and self-checks whether the information received was correct or not.

3. Do-it-yourself evaluation: This is still higher degree of evaluation. This involves more careful planning and applies principles of evaluation and are more systematically done. They usually require surveys or score cards.

Eg: If A.O. gets an information that in an X village hail storm has damaged 100% of crops, he will not totally depend on this information alone but he visits the village and with the help of surveys or score cards, he will evaluate by himself the extent of damage to crops.

4. Extension studies: This is the fourth level of evaluation continuum, complicated than the above three methods. Uses higher tools, techniques, and methods for evaluation purpose. Uses more scientific approach. **Eg:** theses of M Sc and Ph D in extension studies come under this category

5. Scientific research: last on evaluation continuum and most complex. Experimental studies scientifically carried out to determine cause and affect relationships. The scientific research must be

- a. **Factual (or Valid):** Measure what you think you are measuring

- b. **Analytical:** Analyze the relationships of various factors
- c. **Reliable:** Sample representative of population consistency of results.
- d. **Objective:** Free of bias- others get similar results.
- e. **Impartial:** Approach with a open mind and spirit of enquiry

IV. According to the sources, evaluation is divided into 2 types

1. **Internal evaluation:** Internal evaluation can be done in different forms. **Eg: performance appraisals, review committees in an organization etc.**

2. **External evaluation:** When an external person, institution or development agency has done an evaluation work, it is called external evaluation. **Eg:** Evaluation of ATMA project. **Evaluation of human behavior:** Human behaviour can be evaluated with the help of following tests, scales, schedules or questionnaires

Knowledge tests measures extent of information a person has gained because of the programme **Eg-** Test to find out the knowledge gained by the farmers on the SRI method of rice cultivation

Understanding tests: these are also designed to measure the to what extent a person has understood the information

Skill or performance tests or schedules are those which measure the amount of skills acquired by a person. **Eg: Running a conoweeder or marker in SRI rice cultivation Attitude scale:** These are developed to know how people feel or think about certain programmes. For this purpose the person has to mark one of the choices on the scale in terms of degrees of the attitude such as agree, undecided, do not agree etc.

Eg: Scale developed to measure the attitude of the farmers on SRI rice cultivation

Importance of evaluation in Agricultural Extension:

1. Extension evaluation helps to determine the degree to which the important purposes and specific objectives are attained **Eg:** after conducting a training programme on SRI, extension worker can evaluate farmers and can understand whether they gained any knowledge on all principles and practices of SRI cultivation.

2. It also helps to provide periodic tests which gives direction for continuous improvement of work

Eg: If extension worker has delivered lecture on new concept cluster bean cultivation for gum extraction, which is followed by evaluation of training. If majority of the farmers have not understood the concept or technology, lecture training method has to be changed to exposure visit / lecture should be followed by a method demonstration / result demonstration / exposure visit.

3. It helps to serve as a check on extension teaching methods.

4. It helps to furnish data regarding the rural situation to extension programme planning

5. To provide evidence of the value of the programme and a basis for adjusting a programme. **Eg:** If a particular training programme is felt to be of good value by the participants, the same thing can be documented and kept as evidence.

6. To give satisfaction to leaders and cooperators through an understanding of what is accomplished

Eg: Following a method demonstration, if majority of trainee farmers were able to run conoweeder for weeding in SRI paddy; this gives leaders / cooperatives a satisfaction. Good

oral or written performance of farmers for a lecture on marigold cultivation practices, gives a satisfaction not only to extension worker but also for leaders and cooperatives.

7. To help in locating strong or weak points in any programme or plan

Eg: Method demonstration on vermicompost preparation in village could have the following strong and weak points

Strong points	Weak points
Collection of good quality of semi rotten material	Poor gathering
Demonstration done under shade	No wide publicity
Good interaction between farmers and extension worker	Literature not distributed

8. To ascertain the result of organization and administrative procedures of the programme

9. It also helps to establish a bench mark

10. It provides the information with whom we work i.e. people Eg: Evaluation helps to assess certain information about the villagers like their cooperativeness, sincerity, hospitality, hardworking nature, litigant nature, women participation, enthusiasm, risk proneness, impact of various political groups etc.

Chapter 10 COMMUNICATION

Subtopic-I : Meaning & Definition of Communication

Subtopic-II: Purposes of
communications **Subtopic -III:**

Function of communication

Subtopic-IV: Models of
Communication

Subtopic- V: Elements of Communication and their Characteristics

Subtopic-VI: Types of communication

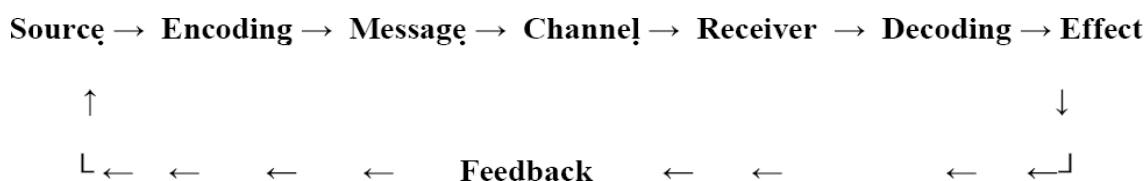
Subtopic-VII: Barrier in communication

Subtopic-I : Meaning & Definition of Communication

The word 'communication' comes from the Latin word *communis*, its meaning is common. It means that when we communicate, we are trying to establish 'commonality' with someone through a message. 'Communication' then is a conscious attempt to establish commonality over some idea, fact, feelings and the like, with others. Consequently it is a process of getting a source and a receiver tuned together for a particular message or a series of messages.

For communicating the ideas, we use a medium. The other man has listened to your concept but before understanding the proper meaning he has done the decoding of that concept in his mind and he has made a decision in his mind for this concept, whether this concept should be used or not, this decision is called the effect of communication. As you are

standing there you are observing the man's reactions and evaluating the "effectiveness of the communication." This evaluation is called "Feedback". We can explain this whole process in the following way:



Definitions of Communication:

Communication is anything that conveys meaning that carries a message from one person to another.

Brooker

(1949). Communication is a process by which two or more persons exchange ideas, facts, or impressions in ways that each gains a common understanding of the meaning, intent and use of message.

-- J. Paul Leagan

(1961). Communication may be defined as "a process by which an individual-the communicator, transmits (usually verbal symbols) to modify the behavior of other individuals - communicates".

Hovland

(1964). "Communication is the control of behavior through descriptive and reinforcing stimuli".

– Hortman

(1966) Communication is any occurrence involving a minimum of four sequential ingredients: (i) a generator of a (ii) sign-symbol system which is (iii) projected to (iv) at least one receiver who assigns meaning.

Robert Goyer

(1967). "Communication has as its central interest those behavioral situations in which a source transmits a message to a receiver(s) with conscious intent to affect the latter's behavior".

– Miller (1968).

Communication is the process of sending and receiving messages through channels which establishes common meanings between a source and a receiver. -Van den Ban and Hawkins(1988).

Communication is a process by which extension workers individually, in a group or through a medium exchange attitudes and share knowledge and or skills on behalf of an organization with farmers/farm women, through such ways that each gains comprehension, understanding and use of the message.

Subtopic-II: Purposes of communications

1. According to Aristotle prime goal of communication is persuasion.
2. According to faculty theory of psychology, purpose of communication is informative i.e. an appeal to the mind.
3. Schramm (1964) described the purpose of communication as immediate reward and a delayed reward.
4. According to Festinger (1957) the purpose of communication as consummators and instrumental.
5. According to Berlo (1960) the sole purpose of communication is to influence.

On the basis of the above descriptions, it can be said that the purpose of communication is to influence – to affect with intent by creating comprehension, understanding and perceived use of the message.

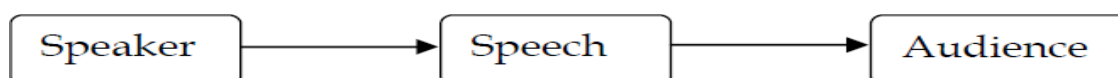
Subtopic –III: Functions of communication

- 1. Information function:** The basic requirement of adapting and adjusting oneself to the environment is information. There must be some information about what is going on in the environment which concerns the people. The getting or giving of information underlies all communication functions, either directly or indirectly.
- 2. Command or instructive function:** Those who are hierarchically superior, in the family, society or organization, often initiate communication either for the purpose of informing their subordinates or for the purpose of telling them, what to do, how to do, when to do etc. The command and instructive functions of communication are more observable in formal organizations than in informal organizations.
- 3. Influence or persuasive function:** According to Berlo (1960), the sole purpose of communication is to influence people. Persuasive function of communication i.e. to induce people is extremely important for extension in changing their behaviour in the desirable direction.
- 4 Integrative function:** A major function of communication is integration or of continuously offsetting any disintegration at the interpersonal or at the organizational level. This helps to maintain individual, societal or organizational stability and identity.

Subtopic-IV: Models of Communication

ARISTOTLE MODEL: According to Aristotle Model, communication process has three elements

1. Speaker – Person who speaks
2. Speech – The speech that the individual produces
3. Audience – The person who listens



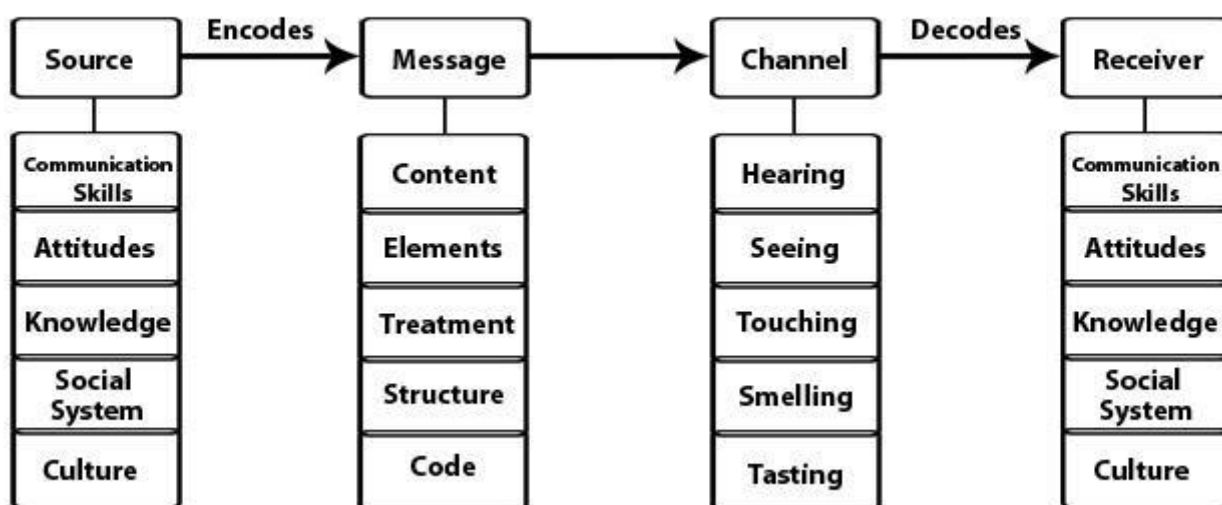
SHANNON-WEAVER MODEL (1949): model is consistent with Aristotle's proposition. According to them, the ingredients of communication are-



Compared with the Aristotelian model, the source is the speaker, the signal is the speech and the destination is the audience, plus two added ingredients, a transmitter which sends out the source's message and a receiver which catches the message for the destination.

BERLO'S MODEL (1960) OF COMMUNICATION PROCESS

Berlos's SMCR Model of communication



The Berlo's model follows the SMCR model. This model is not specific to any particular communication. Berlo's model lives a number of factors under each of the elements:

Source: The source is the message originator.

Communication skills – It is the individual's skill to communicate (ability to read, write, speak, listen etc...)

Attitudes – The attitude towards the audience, subject and towards one self for e.g. for the student the attitude is to learn more and for teachers wants to help teach.

Knowledge – The knowledge about the subject one is going to communicate for e.g. whatever the teacher communicates in the class about the subject so having knowledge in what you are communicating.

Social system – The Social system includes the various aspects in society like values, beliefs, culture, religion and general understanding of society. It is where the communication takes place.

Culture: Culture of the particular society also comes under social system.

Encoder: The sender of the message (message originator) is referred as encoder, so the source is encoding the message here.

Message

Content – The beginning to the end of a message comprises its content for e.g. From beginning to end whatever the class teacher speaks in the class is the content of the message. **Elements** – It includes various things like language, gestures, body language etc, so these are all the elements of the particular message. Content is accompanied by some elements.

Treatment – It refers to the packing of the message. The way in which the message is conveyed or the way in which the message is passed on or deliver it.

Structure- The structure of the message how it is arranged, the way you structure the message into various parts.

Code- The code of the message means how it is sent in what form it could be e.g. language, body language, gestures, music and even culture is a code. Through this you get/give the message or through which the communication takes place or being reached.

Channel- It is nothing but the five senses through this only we do. The following are the five senses which we use

- Hearing
- Seeing
- Touching
- Smelling
- Tasting

Whatever communication we do it is there either of these channels.

Hearing: The use of ears to get the message for e.g. oral messages, interpersonal etc.

Seeing: Visual channels for e.g. TV can be seen and the message is delivered.

Touching: The sense of touch can be used as a channel to communicate for e.g. we touch and buy food, hugging etc.

Smelling: Smell also can be a channel to communicate for e.g. perfumes, food, charred smell communicates something is burning, we can find out about which food is being cooked etc.

Tasting: The tongue also can be used to decipher e.g. Food can be tasted and communication can happen.

Decoder: Who receives the message and decodes it is referred to as decoder.

Receiver: The receiver needs to have all the thinks like the source.

LEAGANS (1963) OF MODEL COMMUNICATION



The task of communication, according to him, is to provide powerful incentives for change. Success at this task requires through understanding of the six elements of communication, a skilful communicator sending useful message through proper channel, effectively treated, to an appropriate audience that responds as desired.

ROGERS AND SHOEMAKER (1971) MODEL OF COMMUNICATION

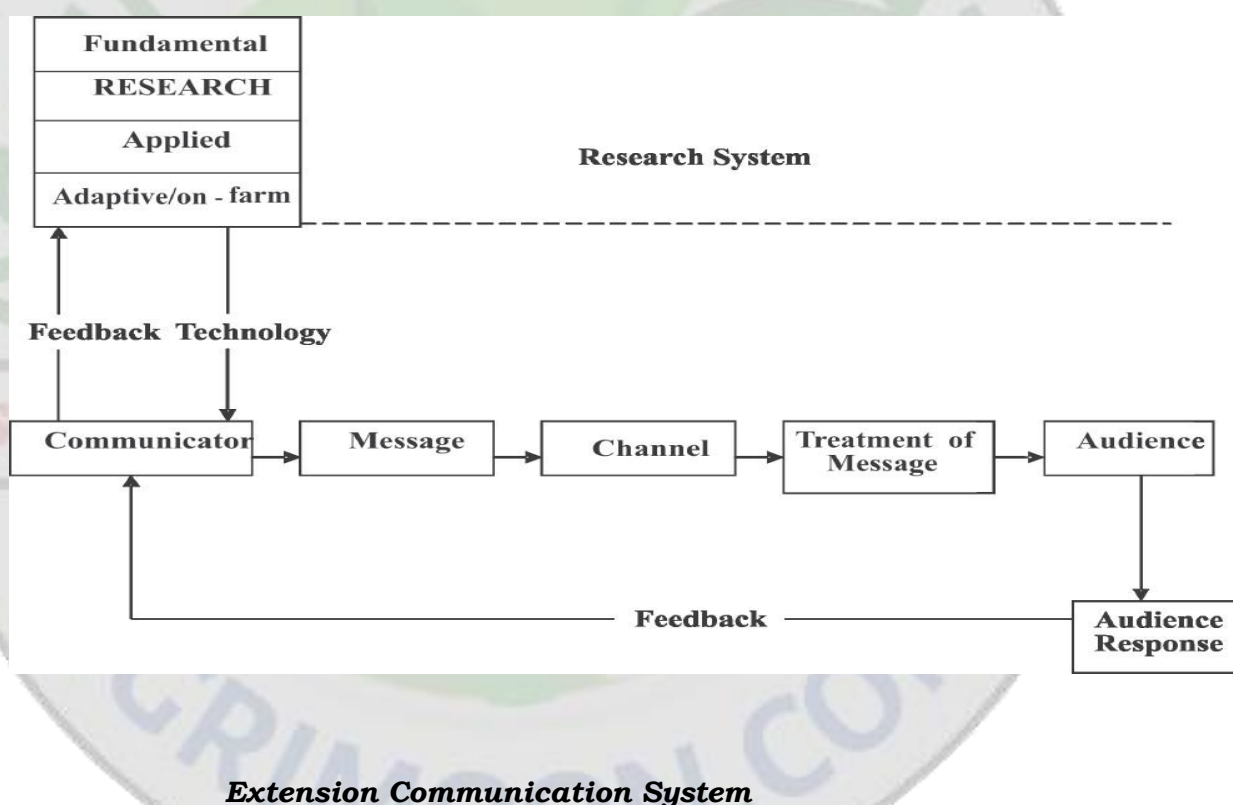


According to them a source (S) sends a message (M) via certain channels (C) to the receiving individual (R), which cause some effects (E) i.e. changing the existing behavior pattern of the receiver.

Communication in extension may also be thought of as two-way stimulus-response situation in which the necessary stimulus is provided by the communicator, the extension agent, in the form of a message, which produces certain response on the audience, the farmers and vice- versa. A favorable response by the audience reinforces learning.

Model on extension communication system

Communication in extension may also be thought of as two-way stimulus – response (S-R) situation in which the necessary stimulus is provided by the communicator, the extension agent, in the form of a message which produces certain response on the audience, the farmers and vice-versa. A favourable response by the audience reinforces learning. A diagrammatic representation of the extension communication system on the basis of the model suggested by Leagans (1963) is presented in the below figure.



Subtopic-V: Elements of Communication and their

Characteristics Successful communication requires a skillful communicator sending a useful message through proper channels, effectively treated to an appropriate audience that responds as desired. The communication task thus consists of the skillful handling of six key elements.

The Communicator:

In the context of agriculture and rural development, extension agent is the communicator who starts the process of communication. The extension agent and mass media like radio are sometimes visualized as sources or originators of messages, which is not correct. Knowledge generates through research and as such the Research Institutes, Universities are the

originators of sources of message. The extension agent obtains the required information from research and carries it to the audience, the farmers. The extension agent is the communicator, a carrier of information. To enhance the process, extension agents may take the help of some aids, known as audio-visual aids. They also carry back the reactions of the farmers, their problems etc. as feedback information to research for finding out solutions for the same.

The characteristics of a good communicator are:

The individual knows:

- i. the objectives - have them specifically defined;
- ii. the audience – their needs, interests, abilities, predisposition;
- iii. the message – its concept, validity, usefulness, importance;
- iv. channels that will reach the audience;
- v. organization and treatment of the message;
- vi. the professional abilities and limitations.

The individual is interested in:

- i. the audience and its welfare;
- ii. the message and how it can help people;
- iii. the results of communication and their evaluation;
- iv. the communication process;
- v. the communication channels – their proper use and limitations;
- vi. improvement of the communication skill.

The individual prepares:

- i. a plan for communication – a teaching plan;
- ii. communication materials and equipments;
- iii. a plan for evaluation of results.

The individual has skill in:

- i. selecting messages;
- ii. treating messages;
- iii. expressing messages – verbal and written;
- iv. the selection and use of channels;
- v. understanding the audience;
- vi. collecting evidence of results.

Poor communicators, on the other hand:

- i. Fail to have ideas to present that are really useful to the audience.
- ii. Fail to give the complete story and show its relationship to people's problems.
- iii. Forget that time and energy are needed to absorb the material presented.
- iv. Feel they are always clearly understood.
- v. Refuse to adjust to closed minds.
- vi. Talk while others are not listening.
- vii. Get far too ahead of audience understanding.
- viii. Fail to recognize others' view-point.
- ix. Fail to recognize that communication is a two-way process.
- x. Let their own biases over-influence the presentation.
- xi. Fail to see that everyone understands questions brought up for discussions.
- xii. Fail to provide a permissive atmosphere.

- xiii. Disregard the values, customs, prejudices and habits of the people, and
- xiv. Fail to start where people are, with respect to knowledge, skill, interest and need.

Message or content:

A message is the information a communicator wishes his audience to receive, understand, accept and act upon. Messages, for example, may consist of statements of scientific facts about agriculture, sanitation or nutrition, description of action being taken by individuals, groups or committees, reasons why certain kinds of action should be taken; or steps necessary in taking given kinds of action. Potential messages range as wide as the content of the programmes is.

Messages related to programmes of change are, therefore, the relevant 'cargo' to be carried to people by the channels of communication. They are the important content, sometimes referred to as 'arguments' 'appeals and 'stimuli'. Whether messages operate effectively as incentives to changed behaviour in any given situation depends on a wide range of influences. A successful communication is one in which the major factors influencing the message are controlled as far as possible. This is the responsibility of the communicator.

A good message must be:

1. In line with the objective to be attained;
2. Clear – understandable by the audience
3. In line with the mental, social, economic and physical capabilities of the audience.
4. Significant – economically, socially or aesthetically to the needs, interests and values of the audience
5. Specific – no irrelevant material;
6. Simply stated covering only one point at a time.
7. Accurate – Scientifically sound, factual and current;
8. Timely – especially when seasonal factors are important and issues current
9. Supported by factual material covering both sides of the argument;
10. Appropriate to the channel selected
11. Appealing and attractive to the audience – having utility, immediate use.
12. Applicable – audience can apply recommendation;
13. Adequate – Combining principle and practice in effective proportion;
14. Manageable – can be handled by the communicator with high professional skill and within the limits imposed by time.

In contrast, poor communicators often –

- i. Fail to clearly separate the key message from the supporting content or subject-matter.
- ii. Fail to prepare and organize their message properly.
- iii. Use inaccurate or fuzzy symbols –words, visuals or real objects – to represent the message.
- iv. Fail to select messages that are in line with the felt needs of the audience.
- v. Fail to present the message objectively – present the material, often biased, to support only one side of the proposition.
- vi. Fail to view the message from the standpoint of the audience, and
- vii. Fail to time the message properly within a presentation or within a total program.

Channel

The sender and the receiver of messages must be connected or '**tuned**' with each other. For this purpose, **channels** of communication are necessary. **Channels** are the physical bridges between the sender and the receiver of messages and the avenues between a communicator and an audience on which messages travel to and fro. They are the **transmission lines** used for carrying messages to their destination. Thus, the channels serve as essential tools of the communicator.

A **channel** may be anything used by a sender of message to connect him with intended receivers. The crucial point is that he must get in contact with his audience. But channels are no good without careful direction or use in the right way, at the right time, to do the right job for the right purpose with the right audience, all in relation to the message.

The channels of communication may be classified into a number of ways according to different criteria.

According to form

Spoken: Farm and home visit, farmer's call, meetings, radio talk etc.

Written: Personal letter, farm publications, newspaper etc.

According to nature of personnel involved

Personal localite: They are the local leaders and local people who belong to the receivers own social system. Personal localite channels are important in traditional social system.

Personal cosmopolite: These are the channels of communication from outside the social system of the receiver. They are the extension agents of various organizations and are important in changing the farmers from traditional to modern.

According to nature of contact with the people

Individual contact: The extension agent communicates with the people individually, maintaining separate identify of each person. Examples are farm and home visit, farmer's call, personal letter etc.

Group contact: The extension agent communicates with the people in groups and not as individual persons. Examples are group meeting, small group training, field day or farmer's day, study tour etc.

Mass contact: The extension agent communicates with a mass of people, without taking into consideration their individual or group identity. Examples are mass meeting, campaign, exhibition, radio, television etc.

Many obstructions can enter channels. These are often referred to as '**noise**' - that is some obstruction that prevents the message from being heard by or carried over clearly to the audience. '**Noise**' emerges from a wide range of **sources** and causes which are as follows;

1. Failure of a channel to reach the intended audience: Usually, no one channel will reach an entire audience. Some examples: Meetings - all people cannot or may not attend. Radio - all people do not have access to a receiving set or may not be tuned in if they did. Written materials - many people cannot read.

2. Failure on the part of a communicator to handle channels skillfully: If in a meeting, tour, radio programme or any other channel is not used according to good procedure and technique, its potential for carrying message is dissipated. For example in a meeting when everyone cannot hear what is said

3. Failure to select channels appropriate to the objective of a communicator: All channels are not equally useful in attaining a specific objective. For example, if an objective was to show a certain group of people how to do something - dig a compost pit, build a sanitary latrine, treat seed, cook vegetables. etc. - the radio; circular letter, or newspaper would not do the job. Obviously, the channel needed is a method demonstration meeting. On the other hand, if an objective was to give general information about subjects like the above or to inform people of events etc., radio, letters and newspapers would be the proper channels to use.

4. Failure to use channels in accordance with the abilities of the audience: Written materials, for example, cannot serve as useful channels for communicating information to people who are unable to read or to understand the level of complexity or abstraction of the message.

5. Failure to avoid physical distraction: When using the channel of meetings, for example, distractions including people moving in and out, loud noises in or out of the group, heat, lighting crowded condition and many other forms of distraction often obstruct successful message - sending. Static on the radio, poor writing, unattractive exhibits are other examples of '**noise**' that lessen the effectiveness of channels.

6. Failure of an audience to listen or look carefully: The only messages that get through to an audience are those which are heard, seen or experienced. An unfortunate tendency of people is not to give undivided attention to the communicator. This is a powerful obstruction that prevents messages from reaching their desired destination.

7. Failure to use enough channels in parallel: The more channels a communicator uses in parallel or at about the same time, the more chances he has for the message getting through and being properly received. No single channel will ordinarily reach all people who need to receive a message. Research indicates that up to five or six channels used in combination are often necessary to get a message through to large numbers of people with enough impact to influence significant changes in behaviour.

8. Use of too many channels in a series: An important principle of communication is that the more channels used in a series the less chance a communicator has for getting his message through to the intended audience.

To help overcome some of the problems of communication, one should take the following factors into account:

- i. The specific objective of the message.
- ii. The nature of the message – degree of directness versus abstractness, level of difficulty, scope, timing etc.
- iii. The audience – size, need, interest, knowledge of the subject etc.
- iv. Channels available that will reach the audience, or parts of it.
- v. How channels can be combined and used in parallel.
- vi. How channels that must be used in a series can be needed to the minimum, and those used made effective.
- vii. Relative cost of channels in relation to anticipated effectiveness.
- viii. Time available to the communicator and to the audience.
- ix. Extent of seeing, hearing or doing that is necessary to get the message through, and
- x. Extent of cumulative effect or impact on the audience necessary to promote action.

Treatment of Messages

Treatment means the way a message is handled, dealt with, so that the information gets across to the audience. It relates to the technique or details of procedure or manner of performance, essential to effective presentation of the message. The purpose of treatment is to make the message clear, understandable and realistic to the audience.

Treatment of the message by the communicator shall depend to a great extent on choice of the channel and the nature of audience. The task cannot be reduced to a formula or recipe. Treatment is creative task that has to be 'tailor-made' for each communication function. For example, treatment of a message will be different when it is conveyed in a meeting, or published in a folder or broadcast. Similarly, there will be difference in treatment of the message according to the level of literacy, socioeconomic status and progressive of the audience.

Designing treatment usually requires original thinking; deep insight into the principles of human behaviour and skill in creating and using refined techniques of message presentation. The following are the three categories of bases useful for varying treatment

Matters of general organization

- i. Repetition or frequency of mention of ideas and concepts.
- ii. Contrast of ideas.
- iii. Chronological – compared to logical and psychological.
- iv. Presenting one side compared to two sides of an issue.
- v. Emotional compared to logical appeals.
- vi. Starting with strong arguments compared to saving them until the end of presentation.
- vii. Inductive compared to deductive.
- viii. Proceeding from the general to the specific and vice-versa and,
- ix. Explicitly drawing conclusions compared to leaving conclusions implicit for the audience to draw.

Matters of speaking and acting:

- i. Limit the scope of presentation to a few basic ideas and to the time allotted – too many ideas at one-time may be confusing.
- ii. Be yourself –you can't be anyone else, strive to be clear, not clever.
- iii. Know the facts- fuzziness means sure death to a message.
- iv. Don't read your speech – people have more respect for a communicator who talks to the audience.
- v. Know the audience – each audience has its own personality, be responsive to it.
- vi. Avoid being condescending (patronizing). Do not talk or act down to people, or over their heads. Good treatment of message results in hitting the target. Never overestimate the knowledge of an audience or underestimate the intelligence.
- vii. Decide on the dramatic effect desired – effective treatment requires sincerity, smoothness, enthusiasm warmth, flexibility and appropriateness of voice, gestures, movements and tempo.
- viii. Use alternative communicators when appropriate, as in group discussion, panels, interviews etc.

ix. Remember that audience appeal is a psychological bridge to getting a message delivered and,

x. Quit on time-communicators who stop when they have 'finished' are rewarded by audience goodwill.

Matters of symbol variation and devices for representing ideas

1. Word symbols- speech 2. Real objects. 3. Models 4. Specimens. 5. Photographs. 6. Graphs.

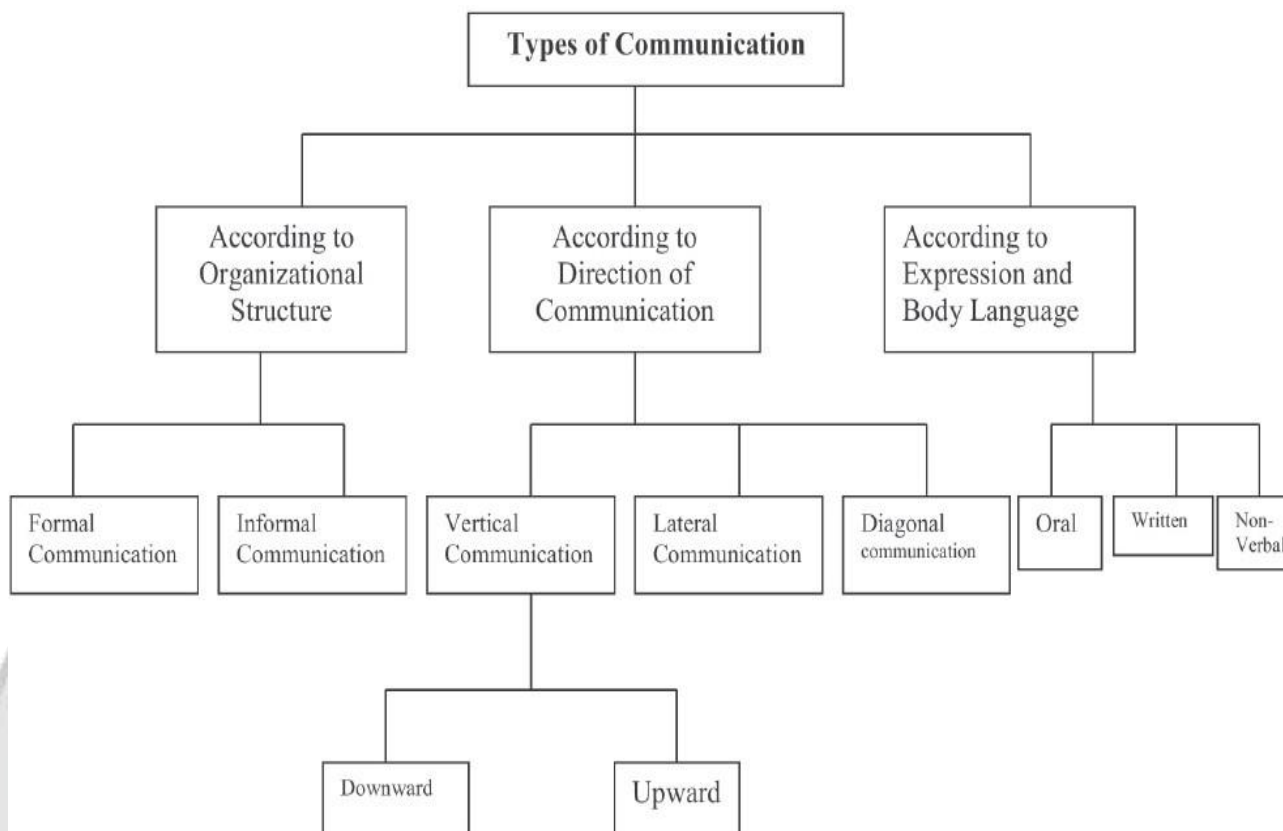
7. Charts 8. Motion pictures. 9. Slides. 10. Drama. 11. Puppets 12. Songs. 13. Flash cards etc.

The Audience

Obviously, an audience is the intended receiver of messages. It is the consumer of messages. It is the intended respondent in message -sending and the assumed to be in a position to gain economically, socially or in other ways by responding to the message in particular ways. In good communication, the audience aimed at is already identified by the communicator. The 'pay off' in communication is dependent on what the audience does in response to messages. An audience may consist of one person or many. It may comprise men, women, or both; youth groups, villagers or their leaders. An audience may be formed according to occupation groups as farmers or artisans; professional groups, as engineers, educators, administrators etc. The importance of clearly identifying an audience cannot be overstressed. The more homogeneous an audience, the greater the chances of successful communication. Likewise, the more a communicator knows about his audience and can pinpoint its characteristics the more likely he is to make an impact. An audience is found by identifying categories such as those previously mentioned. In addition to knowing the identity of an audience and some of its general characteristics, there are other somewhat more specified aspects that help to clarify the exact nature of an audience and how to reach it. The following are some of these:

- 1) Communication channels established by the social organisation.
- 2) The system of values held by the audience - what they think is important.
- 3) Forces influencing group conformity-custom, tradition etc.
- 4) Individual personality factors susceptibility to change etc.
- 5) Native and acquired abilities.
- 6) Educational, economic and social levels.
- 7) Pressure of occupational responsibility-how busy or concerned they are.
- 8) People's needs as they see them, and as the professional communicator sees them.
- 9) Why the audience is in need of changed ways of thinking, feeling and doing.

Subtopic-VI : Types of communication



A) According to organizational structure

1. Formal communication: Such communications are those communications, which are associated with the formal organization structure. The travel through the formal channels - Officially recognized positions in the organization chart. They are established mainly by the organization structure. Formal communications are mostly in black and white. We generally hear the phrase ‘through proper channel’. It explains the essence of formal channels. Such communications include orders, instructions, decisions or intensions etc., of the superior.

2. Informal Communications: Informal Communications are also known as “Grapevine Communications”. They are free from all sorts of formalities, because they are based on the informal relationship between the parties, such as friendship, membership of the same club or association or origin from the same place. Such communication includes comments, suggestions of any other informal reaction also. They may be conveyed by a simple glance, gesture, nod, smile or mere silence too.

B) According to Direction of Communication

Vertical Communication: Upward and downward flow of messages constitutes vertical communication.

1. Downward Communications: Communications which flow from the superiors to subordinates are known as downward Communications. They include orders, rules instructions and policy directives etc. Their nature is directive. It would be impossible to manage an enterprise without downward Communications.

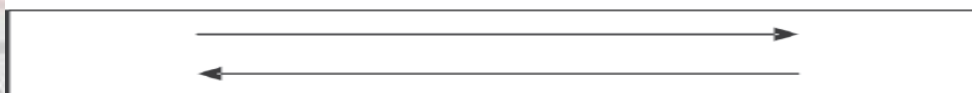
2. Upward Communications: Upward Communications are just reverse of the downward Communications. It flows from the sub-ordinates to their superiors. Such communications include reactions and suggestions from workers, their grievance etc. Contents of the upward Communication are reports, reactions, suggestions, statements and proposals prepared for the submission to the boss etc. Upward Communication is considered to be a main source of motivation in employees.

Exhibit.1. Vertical Communication



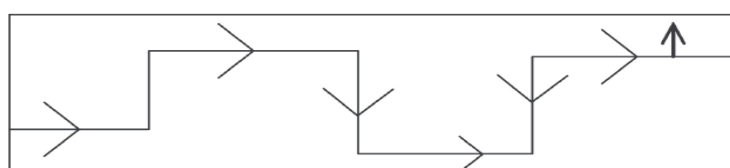
3. Lateral / Horizontal Communication: This type of communication refers to communication between various departments or units representing the same level, or people within the same or different departments, without having a superior – sub-ordinate relationship viz., peer groups, friends and trainees. It flows between persons at the same hierarchical level. The main object of this type of communication is to co-ordinate the efforts of different departments performing different but related activities. Such communications may be oral or written.

Exhibit –2: Lateral Communication



4. Diagonal Communication: In an organization, communication does not necessarily traverse along a stipulated path. While vertical and lateral forms for transmission of messages are important, there is yet another mode to be taken cognizance of, that is, the diagonal. In this type of communications, there is no direct path chalked out for information to travel. It could, at certain stages, take on the upward path, then a lateral direction and finally, move downward, or it could even skip certain stages. This channel proves to be very affective as hierarchical bindings are done away with and communication flows irrespective of position or status. It also helps in building relationships and binding ties between the superior and the subordinate.

Exhibit – 3 Diagonal Communication



c) According to expression and body language

According to way of expression, and body language, the communication may be oral or written and non- Verbal.

1. Oral or verbal communication: In oral communications both parties to the process of communication exchange their ideas through oral words either in face-to face communication or through any mechanical device such as telephone, etc. Meetings and conferences, lectures and interviews are other media of such communications.

2. Written communication: They are communications on black and white. They include written words, graphs, diagrams, pictures etc. They may take the form of circulars, notes, manuals, reports, posters or memos etc.

3. Non-Verbal Communication: Non-Verbal Communication is an integral part of us and helps in communicating effectively. The way an individual positions himself, holds his hands, tilts his head, all transmit volumes about the individual.

Subtopic-VII: Barrier in communication

BARRIERS/ PROBLEMS IN COMMUNICATION

Communication is a process. Process is the act of preceding a series of actions or operations definitely conducting to a desired end. **Each episode of communication has at least three phases:**

- 1) **Expression**
- 2) **Interpretation**
- 3) **Response**

These are the crucial points in communication. If the expression is not clear, the interpretation will be inaccurate and the response improper, thus one's effort to communicate will not succeed. In other words, if the source does not have adequate or clear information if the message is not encoded fully, accurately, effectively in transmittable signs; if these are not transmitted fast enough and accurately enough, despite interference and competition, to the desired receiver; if the message is not decoded in a pattern that corresponds to the encoding; and finally if the destination is unable to handle the decoded message so as to produce the desired response, then, obviously, the system is working at less than top efficiency.

Problems of communication ---

- I. According to phases of communication:
- II. According to various types of problems.
- III. According to nature of problems
- IV. Other classified problems

I. According to phases of communication:

a) RELATING TO THE COMMUNICATOR:

1) **Ineffective environment:** The environment created by the communicator (Extension worker) influences his effectiveness. The physical facilities, air of friendliness, respect of other's point of view, recognition of accomplishments of other, permissiveness and rapport in

general, are all important ingredients of a climate which is conducive to effective communication.

2) **Disorganized efforts to communicate:** to make sense, the communication effort must be organized according to some specific form or pattern.

3) **Standard of correctness:** This involves the use of correct words or other symbols, correct logic and correct content or facts.

4) **Standard of social responsibilities:** This infers that when one communicates, one assumes responsibility for effect of ones communication on the respondents and the society.

5) **Cultural values and social organization:** Cultural values and social organization are determinants of communication. For effective communication, the communicator must possess knowledge of the cultural values of his listeners.

6) **Inaccurate symbols:** The system of symbols used to represent ideas, objects, or concepts must be accurate and used skillfully. The crucial point in the use of symbols to convey ideas is to select those that accurately represent the idea to be conveyed and are understood by the audience. Symbols are meaningful to a person only when he understands what they stand for.

7) **Wrong concept of the communication process:** A common mistake communicated by the communicator is the identification of the part with the whole or the parts fallacy. A successful communication programme of rural development is not a single unit. It requires a series of unit acts. The way one thinks about communication will influence its quality.

b) RELATING TO TRANSMISSION OF MESSAGE:

Many obstructions can enter at the interpretation level. These are often referred to as, 'noise', that is, some obstruction that prevents the message from being heard by or carried over clearly to, the audience. 'Noise' emerges from a wide range of sources and causes which effect the interpretation of the message.

1) **Wrong handling of the channel:** It can be a meeting, tour, radio programme, or one of the other channels, if is not used according to the correct procedure and techniques, its potential for carrying a message is dissipated.

2) **Wrong selection of channels:** All channels are not equally useful in attaining a specific objective. Failure to select channels appropriate to the objective of a communicator will interrupt the interpretation of the message, in the manner in which it is desired, by the intended audience.

3) **Physical distraction:** Failure to avoid physical distraction often obstructs successful message sending.

4) **Use of inadequate channels in parallel:** The more channels a communicator uses in parallel, or at about the same time, the more chances he has of the message getting through and being properly received.

c) RELATING TO THE RECEIVER:

1) **Attention of the listeners:** There is an unfortunate tendency not to give undivided attention to the communicator. This is a powerful obstruction that prevents the message from reaching its desired destination.

2) **Problem of cooperation, participation and involvement:** Both the communicator and the receiver must be brought into the act. Hence, the listener must work a little hard. Learning is an active process on the part of the listener and unless the respondent is on the same wave

length, the character of what is sent out hardly governs the communication process. Thus, it takes two to make communication.

3) **Problem of Homogeneity:** The more homogeneous an audience, the greater the chances of successful communication. Likewise, the more a communicator knows about his audience and can pin-point its characteristics the more likely he is to make an impact.

4) **Attitude of the audience towards the communicator:** An important factor in the effectiveness of communication is the attitude of the audience towards the communicator. It is a function of the communicator to make their attitude favourable. Indirect data on this problem comes from studies of 'prestige' in which subjects are asked to indicate their agreement or disagreement with statements which are attributed to different individuals.

II. According to various types of problems:

These are (1) Technical problems; (2) Semantic problems; and (3) Influential problems.

1) **Technical problems:** These are problems concerned with the **accuracy** of the transference of information from sender to receiver. Certain things that are not intended by the information source are added to the signal. These unwanted additions may be distortions in the shape or shading of a picture or errors in transmission. All these changes in the signal are called 'noise'.

2) **Semantic problems:** Problems regarding the **interpretation of meaning** by the receiver as compared to the intended meaning of the sender. This is a very deep and involved situation even if one is dealing only with the relatively simple problems of communication through speech.

3) **Influential problems:** The problems of influence or effectiveness are concerned with the success with which the meaning is conveyed to the receiver leads to the desired conduct on his part. It may seem, at first glance, undesirably narrow to imply that the purpose of all communication is to influence the conduct of the receiver.

III. According to nature of problems:

1) **Physical problems:** The possible disorders affecting communication fall generally into the following categories. Speech and voice defects; anxiety-tension reaction such as those involved in stage fright, or feeling of inferiority, which noticeably affect speech, paralysis, disease or characteristics of physical appearance which interfere with expressive bodily action or which tend to call forth unfavourable reactions on the part of the listeners; lack of skill in the use of background or staging techniques, together with defects, such as radio station in the means and conditions of transmission.

2) **Psychological:** These psychological difficulties are, in part, a function of the very nature of language; in part, they are due to the emotional characteristics, and mental limitations of human beings. These general considerations concerning the psychological nature of language are the background against which more specific difficulties in communication can be understood. These specific obstacles merit special attention: (i) the failure to refer language to experience and reality, (ii) the inability to transcend personal experience in inter-group communication, (iii) stereotypes, the assimilation of material to familiar frames of reference, (iv) the confusion of precept and concept, ramification and personification.

3) **Cultural:** Cultural differences pose serious barriers in the communication process. Within this expanding field of activity, we may distinguish three small

problems of responsibility raised by our current use of communication systems and (iii) problems of communication when cultural boundaries have to be transcended.

IV. Other classified problems:

- 1) **Entropy and redundancy:** Information is defined in terms of its ability to reduce the uncertainty or **disorganization of a communication situation at the receiving end**. Entropy simply means the uncertainty or disorganization of a system, redundancy is the opposite.
- 2) The idea of noise is another information theory concept which intuitively makes sense in the study of communication. **Noise is anything in the channel other than what the communicator puts there.**
- 3) Error can be reduced as much as desired by keeping the rate of transmission below the total capacity of the channel. If the channel is overloaded, errors increase very swiftly.
- 4) One of the major problems of communication policy and techniques is to find way of controlling the interpretation which an audience will place upon events and notions.

SOME CONCEPTS OF COMMUNICATION:

FRAME OF REFERENCE

Each person has stored experience of beliefs and values as an individual and as a member of the society. This provides the background of stimulation which influences a person's behavior in a particular situation and is called the individual's Frame of Reference.

A message received by an individual is interpreted in terms of the frame of reference of the individual. The message which challenges these beliefs and values may be rejected or misinterpreted. The tendency on the part of the receiver obstructs communication, in case the receiver and the sender do not have a common frame of reference.

PERCEPTION

Gibson (1959) has defined PERCEPTION as the process by which an individual maintains contact with the environment.

Kollar, Blackwell and Engel (1970) explained PERCEPTION as the process whereby an individual receives stimuli through the various senses and interprets them. Perception is influenced by the environment in which communication takes place. It is not the intrinsic quality or attribute of an object, individual or message, but how people individually and collectively perceive them is important for extension.

CREDIBILITY

Credibility is the degree to which a communication source is perceived as trustworthy and competent by the receiver. It means trustworthiness and competence before the audience accepts any message

COMMUNICATION FIDELITY

According to Berlo (1960), Fidelity is the faithful performance of communication process by all its elements: Communicator, message, channel and receiver.

Noise and Fidelity are two sides of the same coin. Eliminating noise increase fidelity, the production of noise reduces fidelity. The communication fidelity finally can be explained as the extent of desirable changes in receivers'

TIME LAG IN COMMUNICATION

LAG means delay.

Communicator shall compute this time lag. The communicator shall plan and initiate the communication action well in advance so that the intended message reaches the audience in time.

COMMUNICATION GAP

Communication gap refers to the difference between what was communicated by the extension agent (communicator) and what has actually been received by the audience.

EMPATHY

Empathy is the ability on the part of one person to understand the other person's frame of mind and reference, and accept the same. This acceptance does not mean agreement.

Empathy can be defined as the ability of an individual to project oneself into the role of another person to be able to appreciate the feelings, thinking and actions of another person. (Rao, 1993)

HOMOPHILY and HETEROPHILY

According to Rogers (1995), a fundamental principle of human communication is that the exchange of ideas occurs most frequently between individuals who are alike, or homophilous.

HOMOPHILY

is the degree to which pair of individuals who interact are similar in certain attributes such as belief, attitudes, values, education, social status and the like.

HETEROPHILY

is the degree to which pairs of individuals who interact are different in certain attributes such as belief, attitudes, values, education, social status and the like.

ENTROPY

Entropy is defined as the ability to reduce the uncertainty or disorganization of a system at the receiving end.

Chapter 11

AGRICULTURAL JOURNALISM

AGRICULTURAL JOURNALISM

According to Webster's Third International Dictionary, Journalism means "the collection and editing of material of current interest for presentation, publication or broadcast". According to Chamber's Twentieth Century Dictionary, Journalism means "the profession of conducting or writing for public journals". The word journalism is derived from "journal"; its best contents are 'dujour' of the day itself. Journal means a daily register or a diary – a book containing each day's business or transactions. The word journal also connotes a newspaper published every day or even less often or a magazine. Thus, journalism means "the collection and editing of material of current interest for presentation, publication or broadcast".

A journal is defined as a register of a diary of public events which has a definite periodicity of publications say a daily, weekly, a bi-weekly, fortnightly and monthly so on.

Journalism is defined as a profession of conducting or writing for a journal which may be a newspaper, a magazine, radio, a television.

Agricultural journalism is journalism as applied to agriculture i.e. Agricultural journalism is a profession of conducting or writing in agriculture and allied subjects for a journal, which may be a newspaper, a magazine, radio, a television.

Importance:

Farmers like many others are also curious. They want to know what is happening or has happened about agriculture. They have a desire for information, because they want to have a better knowledge of the world around them and improved their living standards by increasing the productivity and production. News satisfies this curiosity and this desire for information. People who can read, listen and have information enjoy certain status in our villages. Others look to them as 'knowing' or informed people and look to them for information.

The agricultural extension personnel who are on the job to disseminate the news or transfer of technology to the farming community should invariably understand the agricultural journalism and utilize the mass media channels effectively.

Scope:

The farmers are information hungry and present public extension system is not able to meet the demand of the farmers for information. The farmer and extension worker ratio is widening. On the other side, communication tools development is enormous. Private extension is also coming into picture. Today, journalism in India has got lot of scope with media barons opening new channels or newspapers or publishing houses on a regular basis. The competition is so rife that each channel or newspaper tries to produce something exclusive, which in turn has given the audience a great deal of variety.

Characteristics of News:

- News has geographical boundaries
- News is always revealing
- News is what interests people
- No news interests all people

Factors determining the news values:

1. **Timeliness:** The reader wants his news to be new. News coming from the press must be really not, in the sense of being not only exciting but new, not till they are known. The news must be need based and timely.

2. **Proximity:** Nearness play a dominant role. The reader finds more interest in a minor event closer to his place than a major and important event happened miles away. But proximity is both geographical and emotional. A farmer of nearby village harvesting a record yield is more interesting to the farmers than in other parts of the country or a far off place.

News from research i.e. high yielding variety released by ANGRAU is more important and interests Andhra Pradesh farmers than in Karnataka.

3. **Magnitude (Size):** The very small and very large size also draws the attention of the readers. For example 3000 farmers attending Kisan Mela attracts rather than 100 farmers attending kisan mela. Similarly one or two people died in accident may not be that much important than 40 people died in any accident.

4. **Importance:** This is subjective. It has direct bearing on the kind of audience that the newspaper in mind. What sounds important to a local regional newspaper may not be important to The Hindu and vice-versa.

5. **Truthfulness:** Accuracy of the source of the news is important.

6. **Objectivity:** News reporting should be free from bias. People are interested in the papers that maintain objectivity

7. **Names make news (Prominence):** The important persons like President, Prime Minister, Chief Minister and Governor and they make news rather than ordinary person

8. **Suspense:** Readers are more interested in this type of news which gives some suspense in the beginning and giving the information at the end.

9. **Conflict:** Routine and happiest events may not make news but if any quarrel is there that attracts the readers

10. **Human interest:** Readers are attracted by human interest (names of persons and villages)..

Sources of News:

1. Result demonstrations
2. Research Stations

3. Research publications – Annual reports, highlights
4. Kisan melas
5. Farmers fields
6. Agricultural Universities / State Departments
7. Other extension activities like field days/training programmes, rythu sadassus etc
8. Plan estimates related to agriculture and allied activities
9. Agriculture finance institutions
10. Agriculture input agencies
11. Agriculture Market committees
12. Electricity and irrigation sectors
13. Farmers committees and associations
14. NGOs etc.

Categories or Types of News

- Before and after event stories
- Experience and success stories
- New development – such as pest outbreaks, scientific discoveries, weather and crop conditions, progress made on plans
- Predictions – such as long range crop and livestock report economic outlook information, long range weather forecasts
- Subject matter - when tied to an event, situation, development or problem



Chapter 12

DIFFUSION AND ADOPTION OF INNOVATION

WHY THIS.....

1. There is lag between what is known and what is done by most farmers. Why?
2. Where do most farmers get their new ideas?
3. In some villages, people seem to accept new ideas quickly and in others, nearly all the people are slow to take to new things. Why?
4. Some farmers accept new ideas and put them into practice faster than others. Why?
5. Some new ideas and practices are accepted quickly and with little apparent efforts, while others are accepted only after years of effort put forth by extension agencies. Why?

An extension worker's job does not end with merely informing the farmers about improved practices, he should ensure practical application (by the farmers) of the result of research and field trials. Extension officer's efficiency can be measured

- (a) by the speed or quickness with which the gap between what is known and what is done by the farmers is bridged
- (b) by the number of new practices adopted; and
- (c) also by the number of farmers and communities that adopt the new practices.

Diffusion:

Diffusion is the process by which an innovation is communicated through certain channels over time among the members of a social system. It is a special type of communication, in that the messages are concerned with new ideas. It is this 'newness' of the idea in the message content of communication that gives diffusion its special character. The diffusion of innovations is essentially a social process in which subjectively perceived information about a new idea is communicated.

Diffusion process can be explained with the terms given by Rogers as "the spread of a new idea from its source of invention or creation to its ultimate use of adopters".

Adoption:

Adoption is a decision to make full use of an innovation as the best course of action available. A diffusion of innovation within a social system takes place through its adoption by individual or groups.

The characteristics which determine an innovation's rate of adoption/ or perceived attribute of innovation are:

(1) Relative advantage

Diffusion scholars have found relative advantage to be one of the best predictor of an innovation's rate of adoption. Relative advantage indicates the benefits and the costs resulting from adoption of an innovation. The sub dimensions of relative advantage include the degree of economic profitability, low initial cost, a decrease in discomfort, social prestige, savings in time and effort, and the immediacy of the reward.

(2) Compatibility

Compatibility is the degree to which an innovation is perceived consistent with the existing values, past experiences and needs of the receivers. An idea that is not compatible with the salient characteristics of a social system will not be adopted so rapidly as an idea that is compatible. Compatibility ensures greater security and less risk to the receiver and makes the new idea more meaningful to him. An innovation may be compatible (a) with sociocultural values and beliefs (b) with previously introduced ideas or (c) with client needs for innovations.

(3) Complexity

Complexity is the degree to which an innovation is perceived as relatively difficult to understand and use. Any new idea may be classified on the complexity – simplicity continuum. Some innovations are clear in their meaning to potential adopters others are not. Diffusion of an innovation which is too complex to communicate and to apply is slow.

(4) Trialability

Trialability (divisibility) is the degree to which an innovation may be experimented with on a limited basis before deciding to adopt. New ideas that can be tried on the installment plan will

generally be adopted more rapidly than innovations that are not divisible. Trialability of an innovation is important for its diffusion for several reasons. The feeling of insecurity associated with the adoption of something new and previously unknown is greatly minimized if it can be tried out on a small scale.

(5) Observability to those people within the social system.

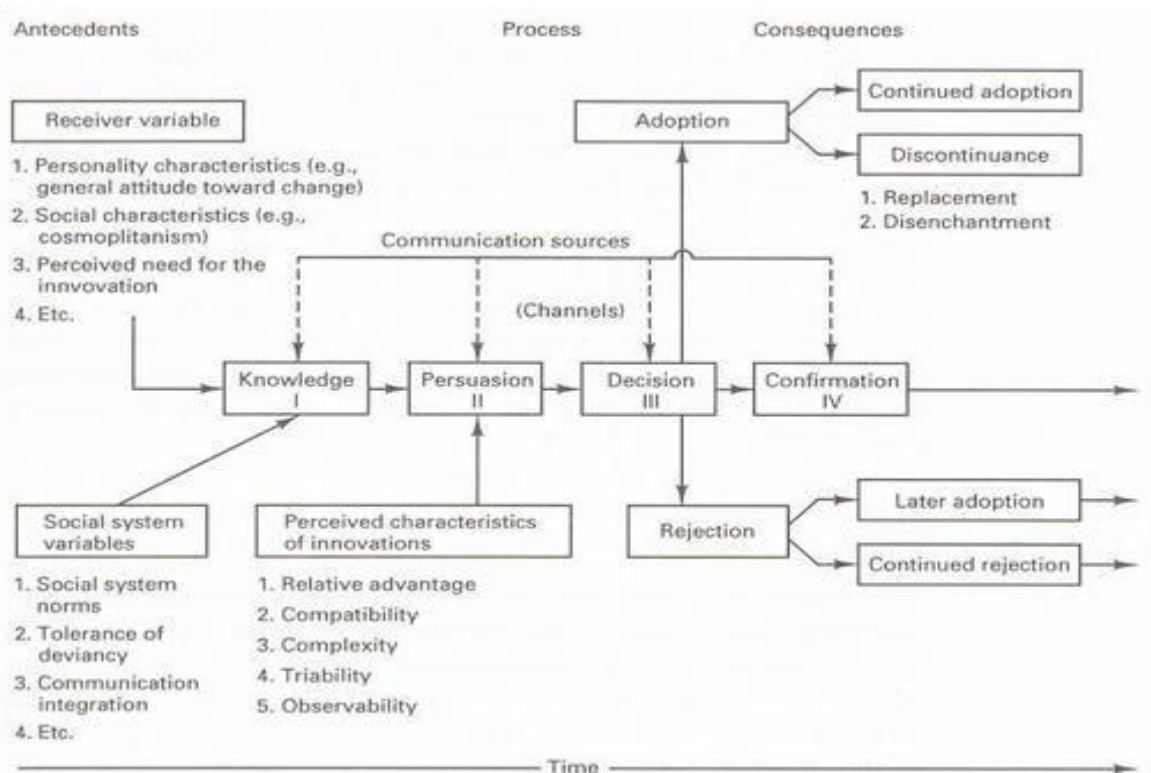
Observability is the degree to which the results of an innovation are visible/observable, demonstrable and communicable to farmers. The results of some ideas are easily observed and communicated to others, whereas some innovations are difficult to describe to others. The observability of an innovation as perceived by members of a social system, is positively related to its rate of adoption. The visible impact of an innovation facilitates its diffusion in the social system.

Meaning of Innovation Decision Process:

The Innovation - Decision process is the process through an individual (or other decision making unit) passes from first knowledge of an innovation, to forming an attitude toward the innovation, to a decision to adopt or reject, to implementation of the new idea, and to confirmation of this decision to adopt or reject and to confirmation of this decision. This process should be distinguished from the diffusion process by which new ideas are communicated

to the members of a social system. The major difference between the two processes is that diffusion occurs among the units in a social system, whereas innovation- decision making takes place within the mind of an individual. The adoption or rejection of an innovation is a decision by an individual. If he adopts, he begins using a new idea, practice, or object and ceases using the idea that the innovation replaces. The innovation-decision is a special type of decision-making; it has certain characteristics not found in other kinds of decision-making situations. In the case of the adoption of an innovation, an individual must choose a new alternative over those previously in existence. Therefore, the newness of the alternative is a distinction of innovation-decision making.

According to Rogers (1983, 1995), the Innovation-Decision Process is the process through which an individual or other decision-making unit passes from first knowledge of an innovation, to forming an attitude toward the innovation, to a decision to adopt or reject, to implementation and use of the new idea, and to confirmation of this decision. This process consists of a series of actions and choices over time through which an individual or an organization evaluates a new idea and decides whether or not to incorporate the new idea into the ongoing system. This behaviour consists essentially of dealing with the uncertainty that is inherently involved in deciding about a new alternative to those previously in existence. The perceived newness is a distinctive aspect of innovation-decision making, compared to other types of decision-making. Innovation-decision is a process that occurs over time and is conceptualized to have five stages as given in below –



Paradigm on Stages in the Innovation-Decision Process

Source: Rogers (1995)

Over adoption

You are well aware that some farmers continue to adopt an innovation, rather vigorously, when experts (Scientists or extension officers) feel that it should not be so done. This is over adoption. An example of this phenomenon is indiscriminate sinking of shallow tube wells in a limited area, which may result in lowering of the water table, ultimately making the irrigation system ineffective.

Excessive use of pesticides is another example of over adoption. Over adoption produces negative effect and may cause distortion or deterioration of the related systems. In sufficient knowledge about an innovation and inability to predict its consequences generally leads to over adoption. As a change agent, your role is to prevent the excessive adoption of the innovation by providing adequate knowledge about the innovation through training and communication.

STAGES OF ADOPTION

The five stages of adoption process are:

- (1) Awareness (2) Interest (3) Evaluation (4) Trial (5) Adoption

Awareness Stage

This is the starting stage wherein the farmer comes to know the existence of the new idea but he doesn't have full information about the idea. At this stage farmer is aware of the idea, but lacks detailed information about it. For instance, the farmers may know SRI cultivation in Rice only the name and may not know what (SRI) is, what it will do and how it will work.

Interest Stage

The farmer develops interest in the innovation and seeks additional information about it either from extension officer or from fellow farmers or from any source, which he feels credible. That means the farmer at the interest stage acquires more information about an innovation or idea. Farmer wants to know, what the innovation/idea is, how it works and what its potentialities are.

Evaluation Stage

The farmer here makes mental application of the new idea in the present and anticipated future situations and decides whether or not to try it. The farmer at this stage judges the utility of the innovation. He/she makes an assessment whether the idea is applicable to own situation and if applied what would be the result. For instance, the farmer after hearing to SRI cultivation in Rice and acquiring more information at the interest stage what are the components and how they improve yield and save water, he/she mentally judge whether SRI cultivation improves rice yields if adopted.

Trial Stage

You are aware that at the first instance, the farmers may not take up any new idea & an innovation right away on a large scale because he/she doesn't want to take risk even though the potential of the idea has been proved. The farmer actually applies the new idea on a small scale in order to determine its utility or feasibility & applicability in own situation. Even though, the farmer takes a decision to try the idea by virtue of its plus points or merits, generally the effectiveness of the idea is tested taking this as small scale trials in their own field standards, even though farmers has thought about it for longtime and gathered information concerning it.

Adoption Stages

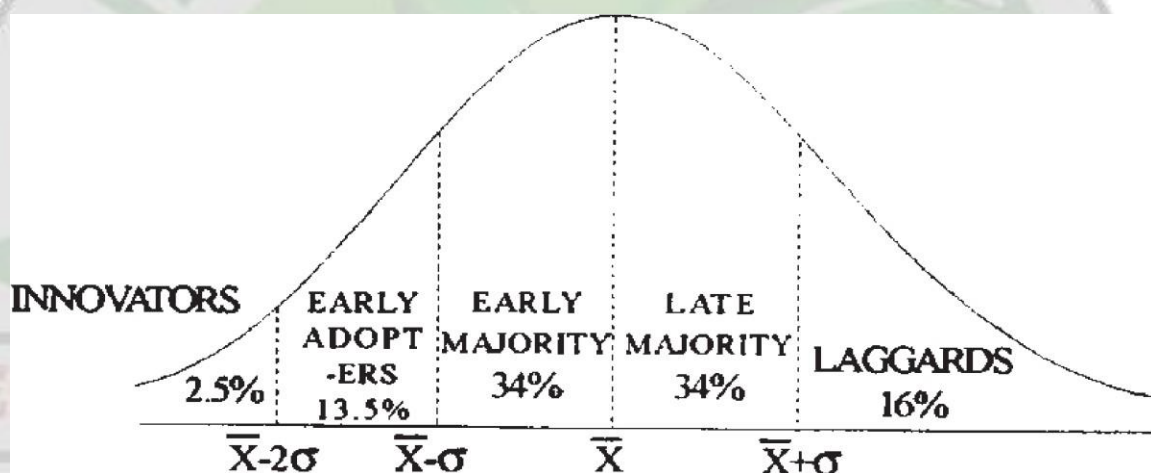
Being satisfied with the performance of the new idea tested on small scale in his own situation, the farmer uses the new idea continuously on a full scale. Trial may be considered as the practical evaluation of an innovation. The innovation becomes a part of his normal farming activity. It provides the advantage of the innovation and hence the farmer takes final decision and applies the innovation in a scale appropriate to own situation on a continued basis.

ADOPTER CATEGORIES

In any social system, when innovation is introduced, there are very few people to immediately accept and put them into use and some farmers take very long time to accept and put into use. All farmers do not adopt at a time. There is a variation among the farmers. Innovativeness is one of the concepts that bring variation among farmers besides other factors. Accordingly farmers can be classified into different categories.

There are different categories of farmers. According to Rogers (1971), the farmers based on their innovativeness can be classified as

1. Innovators (Venturesome)
2. Early adopters (Respectable)
3. Early majority (Deliberate)
4. Late majority (Skeptical)
5. Laggards (Traditional)



Adopter Categories on the basis of innovativeness

1. Innovators (Venturesome)

These are the first people to adopt a new idea, much ahead of other people. They are very few in numbers, probably not more than one or two in a community.

Characteristics:

- a) Have larger farms.
- b) High net worth and risk capital.
- c) Willing to take risks.
- d) Usually not past middle age
- e) Generally well educated
- f) Have respect and prestige in progressive communities but not in conservative type of communities.
- g) Mentally alert and actively seeking new ideas.
- h) Their sphere of influence and activity often goes beyond the community boundaries.

- i) They have many formal and informal contacts outside the immediate locality.
- j) They often by-pass the local extension worker in getting information from the originating sources, and may learn about new things even before he does. They sometimes manage to get samples of seeds or chemicals even before they are released for public use.
- k) They subscribe to many farm magazines and specialised publications.
- l) Other farmers may watch the innovators and know what they are doing but the innovators are not generally named by other farmers as “neighbours and friends” to whom they go for information.

Early Adopter: Respectable

Early adopters are a more integrated part of the local social system than are innovators. Whereas innovators are cosmopolites, early adopters are localities. This adopter's category, more than any other, has the greatest degree of opinion leadership in most social systems. Potential adopters look to early adopters for advice and information about the innovation. The early adopter is considered by many as “the man to check with” before using a new idea. This adopter category is generally sought by change agents to be a local missionary for speeding the diffusion process. Because early adopters are not too far ahead of the average individual in innovativeness, they serve as a role model for many other members of a social system. Members of a social system respect the early adopter. The early adopter is respected by his peers. He is the embodiment of successful and discrete use of new ideas. And the early adopter knows that he must continue to earn this esteem of his colleagues if his position in the social structure is to be maintained.

Characteristics:

- a) Younger than those who have a slower adoption rate, but not necessarily younger than the innovators
- b) They are not the persons who test the untried ideas but they are quickest to use tried ideas in their own situations.
- c) Have large farms.
- d) Higher education than those who adopt more slowly.
- e) High income.
- f) They participate more in the social activities of the community.
- g) They also participate more in government programmes.
- h) This group usually furnishes a disproportionate amount of the formal leadership (elected positions) in the community.
- i) They read papers and farm journals and receive more bulletins than people who adopt later.
- j) They may be regarded as community adoption leaders.

Early Majority: Deliberate (Local Adoption Leaders)

The early majority adopt new ideas just before the average member of a social system. The

early majority interact frequently with their peers, but leadership position; are rarely held by them. The early majority's unique position; between the very early and relatively late to adopt make them an important link in the diffusion process.

The early majority may deliberate for some time before completely adopting a new idea. Their innovation-decision is relatively longer than that of the innovator and the early adopter. "Be not the last to lay the old aside, nor the first by which the new is tried", might be the motto of the early majority. They follow with deliberate willingness in adopting innovations, but seldom lead.

Characteristics:

- a) Slightly above average in age, education and farming experience.
- b) They take a few more farm journals and bulletins than the average.
- c) They have medium high social and economic status.
- d) Less active in formal groups than early adopters, but more active than those adopting later.
- e) In many cases, they are not formal leaders in the association
- f) They also attend extension meetings and farm demonstrations.
- g) They are most likely to be informal resources than early adopters and innovators, and so cannot afford to make hasty or poor decisions.
- h) They associate mainly with people of their own community.
- i) They value highly the opinions their neighbours and friends hold about them; for this is their main source of status and prestige.
- j) They are mostly mentioned as "neighbours and friends" from whom the majority of farmers seek information.

Late Majority: Skeptical

The late majority adopt new ideas just after the average member of a social system. Adoption may be both an economic necessity and the answer to increasing social pressures. Innovations are approached with a skeptical and the late majority do not adopt until most other in their social system have done so. The weight of system norms must definitely favour the innovation before the late majorities are convinced. They can be persuaded of the utility of new ideas, but the pressure of peers is necessary to motivate adoption.

Characteristics:

- a) Those in this group have less education and are older than the early majority.
- b) They form the major part of formal organisational membership, although they participate less in such formal groups.
- c) They take fewer leadership roles than the earlier adopters.
- d) They take and read fewer papers, magazines and bulletins, than the early majority.
- e) They do not participate in as many activities outside the community as do people that adopt earlier.

Laggards: Traditional

Laggards are the last to adopt an innovation. They possess almost no opinion leadership. They are the most localite in their outlook of all adopter categories, many are near isolates. The point of reference for the laggard is the past. Decisions are usually made in terms of what has been done in previous generations. This individual interacts primarily with others who have traditional values. When laggards finally adopt an innovation, it may already have been superseded by another more recent idea which the innovators are already using. Laggards tend to be frankly suspicious of innovations, innovators, and change agents. Their traditional direction slows the innovation decision process to a crawl. Adoption lags far behind knowledge of the idea. Alienation from a too-fast-moving world is apparent in much of the laggard's outlook. While most individuals in a social system are looking to the road of change ahead, the laggards has his attention fixed on the rear-view mirror.

Characteristics:

- a) Least education.
- b) Oldest.
- c) Participate least in formal organisations, cooperatives and government programmes.
- d) They hardly read farm magazines and bulletins.



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